CORE LIFTING

OVERHEAD CRANE HOISTS

R&M MATERIALS HANDLING

CoreLifting.com

R&M MATERIALS HANDLING®

SPACEMASTER® SX WIRE ROPE HOISTS

With nearly a century of expertise, our wire rope hoist products lead the industry in innovation, performance and reliability. We back our products with state-of-the-art proprietary software, support and expertise. Discover why R&M Materials Handling is an ideal choice for your next hoist or crane.

SPACEMASTER® SX WIRE ROPE HOISTS

The Spacemaster® SX represents an innovative design that includes a large drum diameter, giving Spacemaster® SX hoists the lowest headroom in the industry, while reducing rope wear and hook drift.

Standard configurations include normal headroom, low headroom and double girder trolley designs. Hoists in short ton ratings or metric ratings meet a wide variety of application requirements:

- HMI-Certified℠
- 1/2 - 80 ton (500 - 80,000 kg) capacities
- Lifts starting at 13 - 320 ft (4 - 100 m) and beyond
- Lifting speed range 10/1.5 - 62/10 fpm (3/0.5 - 19/3 mpm)
- 208, 230, 460, 575/3/60 or 380/3/50 power supply
- Single-reeved or double-reeved
- Two-component epoxy paint
- CSA C/US rating for electrical enclosures
- Hoisting electrical enclosures with NEMA 3 rating
- Designed to meet and exceed either ASME H3 or ASME H4 duty (FEM 1Am, 2m, or 3m) ratings

LOW HEADROOM TROLLEY

- The best solution for single girder cranes
- Its compact design provides maximum space efficiency

NORMAL HEADROOM TROLLEY

- An ideal selection for monorails and jib crane applications

DOUBLE GIRDER TROLLEY

- Superior clearance and headroom for double girder crane applications

RX PROGRAM
Our most common hoists and options are available through our quick delivery RX Program with a 2-5 day lead time!

THE POWER OF LIFTING

The drive train is the unifying element in R&M's equipment. It´s an integrated package of purpose–built gears, motors and controls, designed and made in–house specifically for cranes and lifting motions. This gives our hoists a longer lifetime and excellent functionality. We aim for the highest levels of safety and reliability in our key components.

CERTIFIED FOR YOUR PEACE OF MIND

The Spacemaster® SX wire rope hoist carries the CSA C/US mark. CSA International is a Nationally Recognized Testing Laboratory (NRTL) accredited by the Occupational Safety and Health Association (OSHA) to test and certify products to U.S., Canadian and International recognized standards.

Certification from CSA International also indicates that the Spacemaster® SX wire rope hoist manufacturing sites conform to a range of compliance measures and are subject to periodic follow-up inspections to verify conformance. These same manufacturing sites are also ISO9001 certified.

Quality assurance with all R&M equipment is very important to us. All equipment is tested locally prior to shipment. Units are shipped with a test certificate, owner's manual and a parts list.

SPECIAL APPLICATION AND HIGH CAPACITY HOISTS

- Main hoist / auxiliary hoist on a common trolley
- Double girder underrunning trolley
- Motorized bottom block
- Specialized controls
- Double girder hoist rotated 90°
- Outdoor use / dusty environment
- High altitude
- Special trolley rail gauges
- Low connection double girder trolleys

The Spacemaster® SX wire rope hoist carries the CSA mark with the indicators "C" and "US". This means that the product has been certified by CSA International for both the U.S. and Canadian markets, to the applicable U.S. and Canadian standards. CSA International is a Nationally Recognized Testing Laboratory (NRTL) accredited by the Occupational Safety and Health Association (OSHA) to test and certify products to U.S., Canadian and International recognized standards.

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BUILT FOR YOUR APPLICATION

R&M's product line offers the broadest product portfolio available in the industry. Our extensive line of innovative features allow us to cover nearly every hoisting application and environment. Each hoist is tailored specifically to your needs bringing you the level of quality engineering you rely on with R&M.

PRECISION LOAD HANDLING

Inverter controls, sometimes called variable frequency drives (VFD) are an electrical device which controls the speed of the motor.
ControlMaster® inverter controls provide better performance and prolong the equipment's life. Dynamic braking (a technique of electric braking in which the speed reduction is controlled by the inverter) means the inverter controls the motor speed to nearly zero before the brake is engaged, significantly reducing brake wear. Controlled acceleration and deceleration provides smooth starts and stops and lowers starting currents, reducing gear and motor wear and structural stress on mechanical components and reducing load swing- leading to longer equipment life and better safety.

SAFETY AT YOUR FINGERTIPS

Hoist monitoring is a standard feature on every Spacemaster SX hoist. Hoist monitoring protects your equipment from overloads while providing you with accurate real-time information about how your equipment is being used. We have used our nearly 90 years of experience in the hoist and crane business to also build in helpful service warnings to help you plan your routine maintenance. Monitoring functions do not interfere with the use of your hoist; however, in the case of an overload or sudden loading change, loads are able to be safely lowered to the ground. The HoistMonitor Enclave provides wireless connectivity with the OLI app so you can instantly retrieve real-time HoistMonitor data from the floor without any downtime.

DIGITAL TOOLS & INNOVATION

R&M recognizes the important role digital tools play in the crane and hoist industry and works to rise above competition with innovative technology. Digital tools such as the Overhead Lifting Information (OLI) app and TD Drive Tool app make it easier than ever to stay ahead of competitors and maintain a safe work environment.

The OLI app provides wireless real-time access to raw data without any downtime in a more user-friendly interface. The TD Drive Tool app allows you to display all parameters available (output frequency, active fault codes and more) – right from your mobile device.

ROPE DRUM DESIGN

To achieve maximum performance while reducing hook drift, R&M offers single-reeved hoists with a wide rope drum diameter design.

The wide diameter rope drum reduces hook drift to just 1/8th of an inch per vertical foot of lift, giving operators near-true-vertical lifting accuracy, especially when we consider that most setting adjustments are happening within the last few inches of a lift.

The R&M Spacemaster® SX hoist's wide drum diameter allows for greater lifting heights and more precise operations within a compact, space-saving design. Because R&M's wide diameter rope drum is more compact, end approach is also improved – meaning more workable floor space compared to cranes with hoists that have long, narrow diameter drums.

In most scenarios, R&M's single-reeved hoist will cover all of your application's needs. If 1/8th of an inch of hook drift per vertical foot of lift is still not precise enough, additional features can be added to make an even further impact on efficiency and productivity.

SINGLE-REEVED HOIST

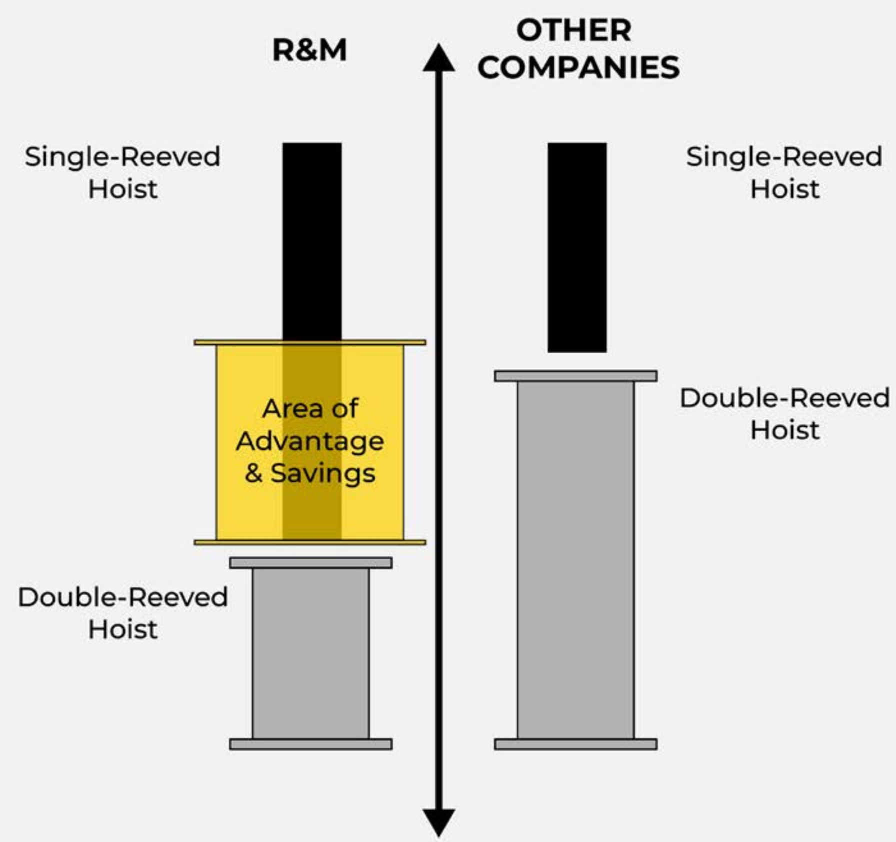
- Lower Up-Front Cost
- Lower Maintenance Costs
- More Compact/Better Floor Coverage

DOUBLE-REEVED HOIST

- Greater Precision
- Higher Capacities

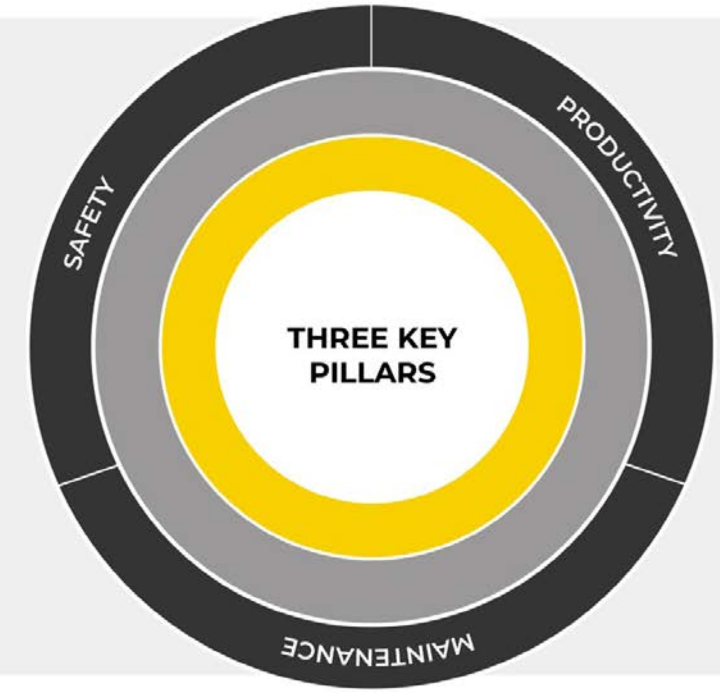
ADDITIONAL R&M FEATURES

- Intelligent Controls
- Automation
- Closed-Loop Inverters
- Micro-Speeds



THREE KEY PILLARS OF TOTAL COST OF OWNERSHIP AND LIFETIME VALUE

When safety, productivity, and preventive maintenance are all working together, your manufacturing business can experience a range of benefits that contribute to lower TCO. Fewer accidents and injuries result in lower costs, increased productivity leads to greater efficiency, and preventive maintenance helps to reduce downtime and prolongs the life of your hoists and cranes.



THREE KEY PILLARS OF LIFETIME VALUE

Understanding the overall cost of your equipment benefits your project by producing a better return on investment (ROI) and improved performance. Keeping TCO as low as possible is the reason so many businesses fret over the dreaded "repair-or-replace" decisions when it is time to replace failing equipment. R&M's cranes and hoists are built with three key pillars -- safety, productivity and maintenance -- in mind to ensure a lower total cost of ownership.

1. Safety: By prioritizing safety, you can protect your employees and equipment, reduce expenses and increase profitability in the long run.
2. Productivity: Preventative maintenance and regular inspections can reduce the risk of unexpected downtime and minimize the associated costs, improving a crane's overall TCO and lifetime value.
3. Maintenance: Keeping a crane well-maintained can also extend its working life, providing greater value over time and reducing the need for premature replacement, thereby maximizing your return on investment.

QX® MODULAR CRANE PACKAGES

QX® modular crane packages are pre-engineered, complete crane component packages utilizing cutting edge technology. Variable frequency drives are standard on trolley and bridge motions for smooth starts and stops. Plug-in cabling* and bolted structural connections simplify assembly. QX® modular crane packages are equipped with your choice of a C-track festoon system or the NRGmaster electrification system.

STANDARD FEATURES

- Spacemaster® SX wire rope hoist
- Top or underrunning design
- PRQ ergonomic pendant controller
- Optional RaCon® 3 radio controller
- HoistMonitor® or HoistWatch load monitoring system
- Standard inverter controls for trolley and bridge
- Plug-in connections for easy assembly and maintenance*
- 208, 230, 460 or 575/3/60 power supplies, as well as 380/3/50
- All components are CSA C/US approved

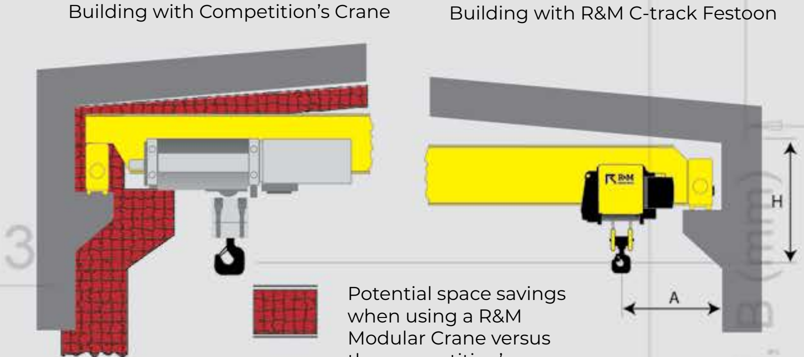
C-TRACK FESTOON SYSTEM

- PVC flat cables are UL & CSA C/US listed
- Suspended from galvanized steel C-track
- All crane motions can be controlled from a convenient plug-in push button station
- Independent C-track for pendant station allows operator to be positioned at a safe distance from load

SPACEMASTER SX CLEARANCES =
SPACE SAVINGS IN NEW CONSTRUCTIONS

Unmatched clearances - Lower operating cost - Maximum flexibility

Building with Competition's Crane — Building with R&M C-track Festoon



Potential space savings when using a R&M Modular Crane versus the competition's crane

A: Distance from hook to wall
H: Distance from hook to ceiling

* Plug-in connections not available for some environments and features.

NRGMASTER ELECTRIFICATION SYSTEM

The NRGmaster energy chain crane features our innovative NRGmaster festoon and RaCon® 3 radio controller for the best floor coverage and ease of operation. The NRGmaster energy chain crane helps you to achieve the best crane clearances and shop floor coverage, while reducing the maintenance concerns associated with traditional C-track festoons.

NRGmaster Electrification System = The Most Space Savings in New Construction

- Unmatched clearances – The floor area under your new crane can now be used more efficiently – Better hook to wall dimensions (A) compared to the competition
- Lower roof heights/maximum lifting heights – Your new building design can be smaller and more functional giving you savings in initial construction. Better hook to ceiling dimensions (H) compared to the competition
- Lower operating costs – Continual savings in heating and/or air conditioning expenses over the lifetime of your building
- Maximum flexibility – Factory layout and material flow planning is optimized

Figure 1. C-track Festoon

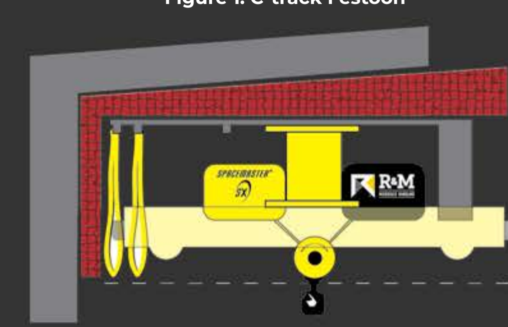
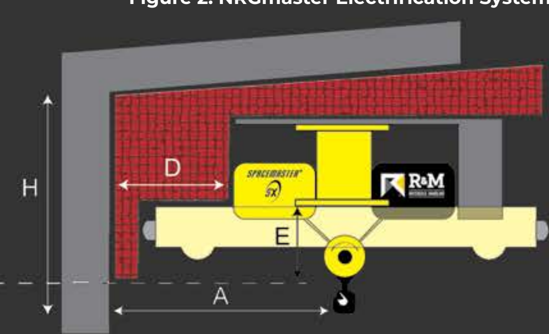


Figure 2. NRGmaster Electrification System



Potential space savings when using R&M's QX modular crane package versus competitors' cranes (Figure 1) and the added space savings versus standard C-track festoon system cranes (Figure 2)

A. Decreased distance from hook to wall (versus C-track festoon cranes)
D. Increased inner span distance to wall (no festoon obstruction)
E. Less obstructions from floor to girder (no dangling cables)
H. Decreased distance from hook to ceiling (versus competitors cranes)

RISE ABOVE WITH R&M MATERIALS HANDLING, INC.

R&M has been innovating since 1929. With our heritage of innovation, our wire rope and chain hoist products have consistently set the crane industry norms. Our cutting-edge solutions will help you to rise above your material handling challenges. You deserve the best performance possible; let us help you achieve it!

R&M Materials Handling, Inc. may alter or amend the technical specifications identified herein at any time with or without notice.
R&M Materials Handling, Inc. is a proud member of the following organizations:

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R&M
MATERIALS HANDLING

SPACEMASTER® SXR ROPE HOISTS

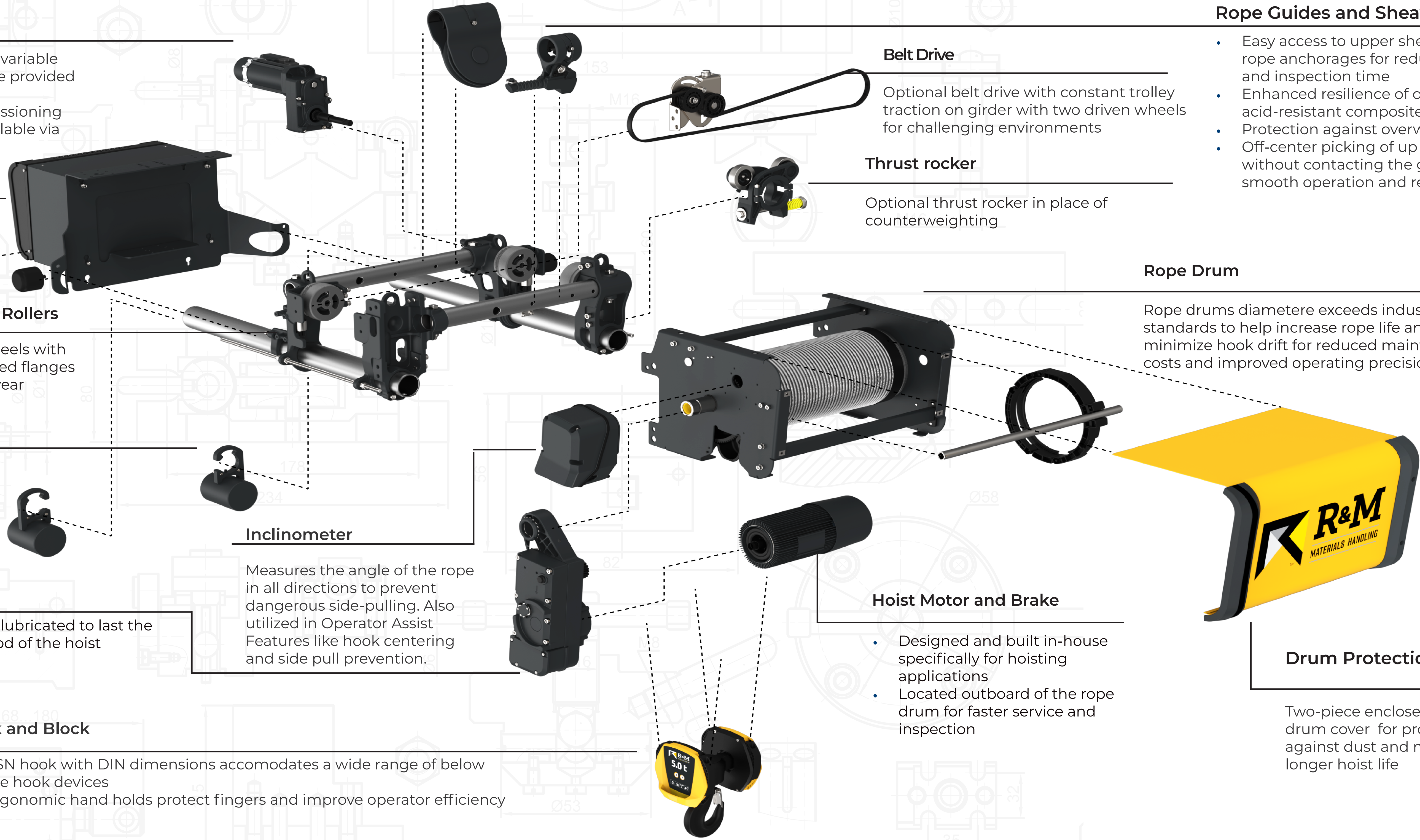
R&M's Spacemaster SXR rope hoist combines the powerful features of its predecessor and the cutting-edge technology of the future. A newly revamped design and set of features coupled with the industry-leading space savings and headroom you've come to expect from R&M.

RISE ABOVE™

RISE TO THE CHALLENGE OF YOUR NEXT PROJECT

SXR Rope Hoist

GROW WITH THE FUTURE OF LIFTING TECHNOLOGY



Trolley Travel Motion

- Inverter controls for infinitely variable trolley travel speed control are provided as a part of standard offering
- Inverter performance, commissioning and troubleshooting info available via easy-to-use smart phone app

Hoist Panel

All controls including the ControlMaster® Edge with W motor, are housed in one singular panel

Trolley Wheels with Guide Rollers

Universal flangeless trolley wheels with guide rollers to fit flat or tapered flanges while reducing girder beam wear

Counterweights

Standard for hoist balancing across a wide range of trolley beam widths

Hoist Motor Gearbox

Permanently sealed housing, lubricated to last the entire designed working period of the hoist

Inclinometer

Measures the angle of the rope in all directions to prevent dangerous side-pulling. Also utilized in Operator Assist Features like hook centering and side pull prevention.

Hook and Block

- RSN hook with DIN dimensions accomodates a wide range of below the hook devices
- Ergonomic hand holds protect fingers and improve operator efficiency

Belt Drive

Optional belt drive with constant trolley traction on girder with two driven wheels for challenging environments

Thrust rocker

Optional thrust rocker in place of counterweighting

Rope Guides and Sheaves

- Easy access to upper sheaves and rope anchorages for reduced repair and inspection time
- Enhanced resilience of durable, acid-resistant composite material
- Protection against overwrapping
- Off-center picking of up to 4° without contacting the guide for smooth operation and reduced wear

Rope Drum

Rope drums diametere exceeds industry standards to help increase rope life and minimize hook drift for reduced maintenance costs and improved operating precision

Hoist Motor and Brake

- Designed and built in-house specifically for hoisting applications
- Located outboard of the rope drum for faster service and inspection

Drum Protection

Two-piece enclosed rope drum cover for protection against dust and moisture for longer hoist life

INNOVATIVE LIFTING SOLUTIONS FOR SAFE AND FORWARD-LOOKING OPERATIONS

R&M is proud to introduce the SXR hoist to the well-known Spacemaster line. This innovation in lifting technology combines industry-leading features with key enhancements that address the feedback we've heard about its predecessor. This versatile hoist integrates advanced controls, integrated communication via the OLI app and visibility across the entire crane system. Designed with ease of maintenance in mind, this hoist retains the reliable cooling and insulation you trust, the SXR enables cost-effective, high-performance operation for applications across industries and environments.

TECHNICAL FEATURES AND OPTIONS

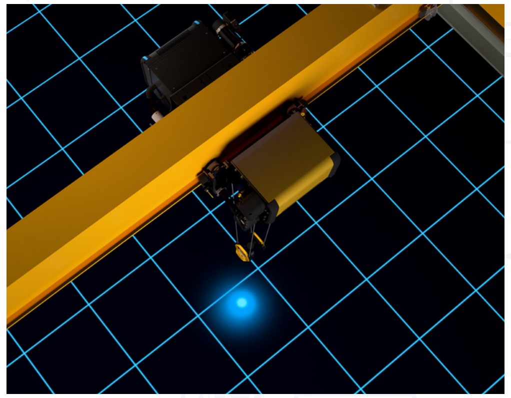
Enhanced protection for operators and equipment

Condition Monitoring

The HoistMonitor® Enclave condition monitoring device is standard with the Spacemaster SXR hoist, and it gives you up-to-date operating data and evaluations throughout the SXR's operating cycle. Full transparency on utilization, usage, risk of downtimes and service life means facilities can make data-driven maintenance decisions.

Blue Safety Light

Promote greater operator safety with the SXR's load spotting light. A blue dot of light is projected on the floor to indicate the hook position, ensuring that everyone on the workfloor has clear line of sight to the hook and load's positioning. This allows operators to predict hook placement before attaching it to the load, which is especially useful in tight spaces where maneuvering the load is difficult.

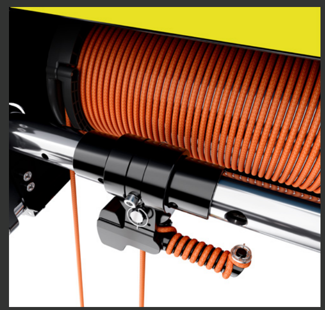


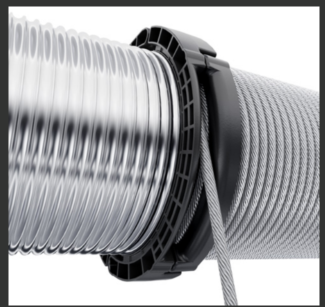
YOUR LIFT - YOUR ROPE - YOUR CHOICE

Choose between the industry-standard steel wire ropes or the premium, lower-maintenance synthetic rope. Although new to the overhead crane industry, synthetic ropes have been used successfully in harsh and corrosive environments, such as maritime environments, for decades.

About Synthetic Ropes

Crafted from high-modulus polyethylene fibers and 85% lighter than steel, synthetic rope suits indoor and outdoor-under-roof environments, operates efficiently within a temperature range of 14°F to 113°F (-10°C to 45°C) and requires no lubrication. Synthetic ropes also offer expanded protection beyond steel wire rope in some corrosive environments, without the reduced breaking strength associated with stainless steel rope.





OLI Comms+: CONNECTED FEATURES

Faster cycle times. Improved operator efficiency. Greater workplace safety.

Control and monitor all critical functions with SXR's OLI Comms+ protocols:

SXR's OLI Comms+ uses standard industrial communication protocol to create a network where all devices on the crane communicate seamlessly with one another for all crane operations. This unified network enables powerful smart features while simplifying troubleshooting and giving a holistic view of the crane's health.

Anti-Sway Technology

Prevent and correct any load sway with active counter-movements of the crane. Sway correction is based on active rope angle measurement.

Follow-Me*

Position the crane over the load by guiding the hook by hand for reduced cycle time.

Slack Rope Protection

Continuously monitor rope tension: automatic cut-off of the hoist motor when slack rope is detected.

Hook Centering*

Automatically positions the crane hook directly above the load. Faster load cycles and improved ease of operation.

Load Restricted Area

Set defined areas of the factory floor where the crane can only be operated below the rated capacity.

Side Pull Prevention*

Rope angle monitoring automatically prevents lifting if rope angle exceeds 4° from vertical, to prevent damage to critical lifting components.

Restricted Zones

Define restricted zones where the hoist may not enter, for example, to avoid machinery or structures that are on the factory floor.

Tandem Control

Electronically synchronized lifting across two hoists for precisely lifting a single load in tandem.

* This Operator Assist Feature requires a full fieldbus crane to activate and utilize

Further options for customized configuration:

- Radio control
- Second hoist brake for SXR 5 & 10
- Synthetic rope
- Ramshorn hook
- IP66 protection
- Stand-by heating
- Large load display
- Rain covers
- Limit switches
- Anti-collision protection
- Rail sweeps
- And more

STAY CONNECTED WITH OLI PRO FOR SXR

Hoist Monitoring, Communication and an Ecosystem of Data

Limited access to your data is a thing of the past with the OLI App and its advanced monitoring system: OLI Comms.

- Access performance data in real-time
- Use real diagnostics to predict maintenance needs
- Address faults and safety concerns proactively
- Reduce maintenance time and visits
- Plan for service downtime in advance to minimize costs

Which OLI is Right for You?

OLI App	Free app available on iOS and Google Play stores. App is able to be used with HoistMontior Enclave devices, when the devices are equipped with the approprate hardware.
OLI Comms	Combination of the HoistMonitor Enclave and gateway modem devices provided with the equipment, which facilitates the usage of all OLI digital services, including OLI App+ and OLI Pro.
OLI Comms+	Uses a communication protocol to gather data from HoistMonitor Enclave and ControlMaster inverters on the crane. This creates an ecosystem of information that improves your ability to predict maintenance needs and indicate areas for performance improvement. OLI Comms+ facilitates the usage of all OLI digital services and enables usage of connected features.
OLI App+	Additional subscription within the OLI App, which allows technicians to edit parameters via the mobile app while connected to the HoistMonitor Enclave device via WiFi or HoistMonitor device via Bluetooth.
OLI Pro	Additional subscription within the OLI App, which allows servuce companies to remotely gather and view HoistMonitor Enclave data offside through cloud services.

INVERTER CONTROLS COMPLETE THE OLI COMMS ECOSYSTEM

OLI Comms uses a communication protocol to gather data from all the electronic devices on your SXR crane. This creates an ecosystem of information that improves your ability to predict maintenance needs and indicate areas for performance improvement. SXR is available with a wide range of inverter and speed options to meet your needs.

Trolley and Bridge Travel Controls:

Inverter controls are R&M's standard for trolley and bridge travel motions. Inverter controls reduce load swing and eliminate brake wear.

Hoisting Controls:

Two-speed hoisting controls are the standard for the SXR. ControlMaster Edge hoisting inverters offer extended speed ranges and reduced brake wear. ControlMaster Elite hoisting inverter controls offer the same advantages as the Edge inverters, plus expanded Operator Assist Features and high-precision speed control.

Why choose R&M's Spacemaster SXR?

- Maintenance-friendly, robust mechanical design
- Data-driven maintenance with HoistMonitor Enclave and OLI Comm network
- Connected features support operator efficiency and safety
- Core of lifting, drive train components are designed and built in-house, specifically for hoisting applications
- Options for tough environments, like synthetic rope, thrust rocker and belt drive

Frame	Reeving*	Load Capacity [ton]									Lifting Height	Hoisting Speed**	Configuration***			
		1.0	1.5	2.0	2.5	3.0	5.0	7.5	10	12.5	[ft]	[fpm]	FO	LO	NO	DO
3 ton	021	ASME H4*	ASME H4								42'-7"; 59'-0"	40 / 6.6		■	■	
	041				ASME H4*	ASME H4					21'-3"; 29'-6"	20 /3.3		■	■	
5 ton	021			ASME H4*	ASME H4						42'-7"; 59'-0"; 78'-8"	40 / 6.6	■	■	■	■
	041						ASME H4				21'-3"; 29'-6"; 39'-4"	20 / 3.3	■	■	■	■
10 ton	021						ASME H4				59'-0"; 78'-8"; 98'-5"	40 / 6.6	■	■	■	■
	041							ASME H4*	ASME H4		29'-6"; 39'-4"; 49'-2"	20 / 3.3	■	■	■	■
	041									ASME H3		16 / 2.5	■	■	■	■

* 021 = two part single reeved / 041 = four part single reeved ** Standard speed with two-speed contactor control
*** F0 = foot mount / L0 = low headroom / N0 = normal headroom / D0 = double girder

All components are CSA C/US approved

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RISE ABOVE WITH R&M MATERIALS HANDLING, INC.

R&M has been innovating since 1929. With our heritage of innovation, our wire rope and chain hoist products have consistently set the crane industry norms. Our cutting-edge solutions will help you to rise above your material handling challenges. You deserve the best performance possible; let us help you achieve it!

R&M Materials Handling, Inc. may alter or amend the technical specifications identified herein at any time with or without notice.

R&M Materials Handling, Inc. is a proud member of the following organizations:

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SPACEMASTER® EX HAZARDOUS LOCATION HOISTS

The Spacemaster EX electric wire rope hoist is the hazardous location version of the industry leading Spacemaster SX hoist. Along with exceptional hook approaches and heights of lifts, this hoist features one of the lowest headrooms in the industry. The Spacemaster EX hoists are rated for use in Class I, Division 2 environments with Groups B, C and D, and Temperature Class T3 or in Class I, Division 1 environments with Groups C and D and Temperature Class T4.

RISE ABOVE™

CSA mark shown with adjacent indicators "C" and "US" for Canada and U.S. indicates that the products have been manufactured to the requirements of both Canadian and U.S. Standards

SPACEMASTER® EX CLASS I, DIVISION 2 HOISTS

The Spacemaster EX Class I, Division 2 hoist carries a Class I, Division 2 hazard location rating for Groups B, C and D and with a temperature class T3, and carries an additional marking of Class I, Zone 2 IIB+H2, T3 . Class and Division rated equipment can be used in the appropriate Zone environment as permitted by the National Electrical Code (NEC 505.9 and NEC 505.20) and the Canadian Electrical Code (CEC C22.1).

CID2 STANDARD FEATURES:

- 1 - 80 ton (1000 - 80,000 kg) capacities
- 460, 575/3/60 Hz or 380/3/50 Hz supply
- Two-speed hoist with 6:1 ratio, contactor control
- Two-speed trolley with 4:1 ratio, contactor control
- Low headroom, normal headroom and double girder trolleys
- Single reeved or double reeved
- Spark-resistant wire rope as standard (EN 13463-5)
- CSA C/US approved

CLASS I, DIVISION 1 vs. CLASS I, DIVISION 2

There are many features that make the Spacemaster EX CID1 hoist different than the CID2 hoist, including:

- Different hoisting and traveling motors
- Bigger hoist control panel and new panel fixing plates
- Armored cables instead of flexible conduits for motors
- Different cable entries for motor cables due to armored cables
- Intrinsically safe circuit for control commands and limit switches
- Additional barrier relays because of intrinsically safe circuit
- Additional labels to mark intrinsically safe circuit cables outside the enclosure
- Stainless steel hoist data and warning plates

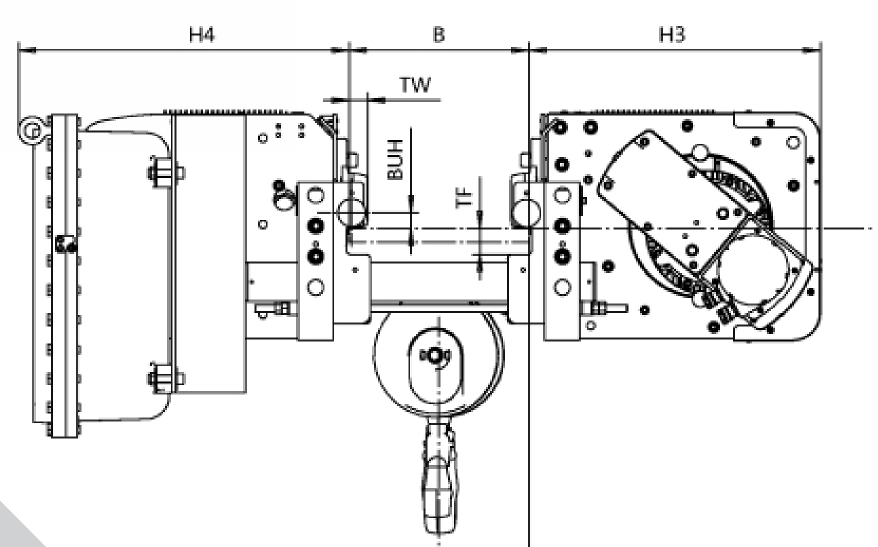
NFPA 70 NEC hazardous location CLASS I *means flammable gasses or vapors may be present ...*	
DIVISION 1	• under normal operating conditions. • as a result of frequent maintenance or repair operations. • as a result of equipment breakdown, faulty operation or failure.
DIVISION 2	• and are normally confined within closed containers when handled or used. • and are normally protected by positive mechanical ventilation. • and the location is adjacent to a Class I, Division 1 location.

What does intrinsically safe-wired mean?

Intrinsically safe electrical systems are designed to limit the electrical and thermal energy in a circuit to a level below what is required to ignite an explosive gas or dust atmosphere. A circuit design considers potential energy that may be present in field devices and interconnecting wiring. Intrinsically safe circuits are used for control commands and limit switches on CID1 equipment.

SPACEMASTER® EX CLASS I, DIVISION 1 HOISTS

The Spacemaster EX Class I, Division hazardous location hoist is certified to meet Class I, Division 1, Groups C & D, and T4 temperature class, and also carries an additional marking of Class I, Zone 1, IIB, T4

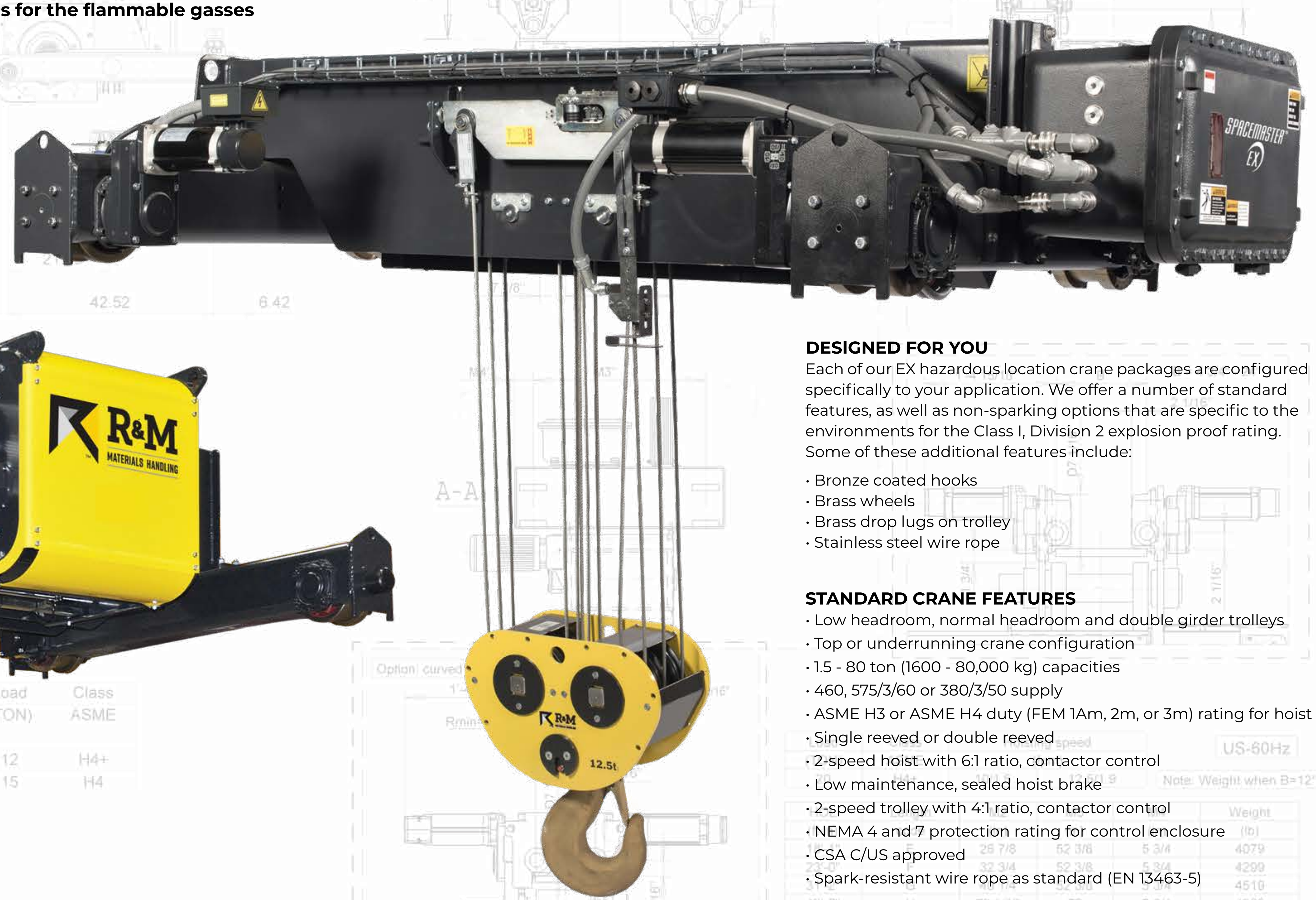


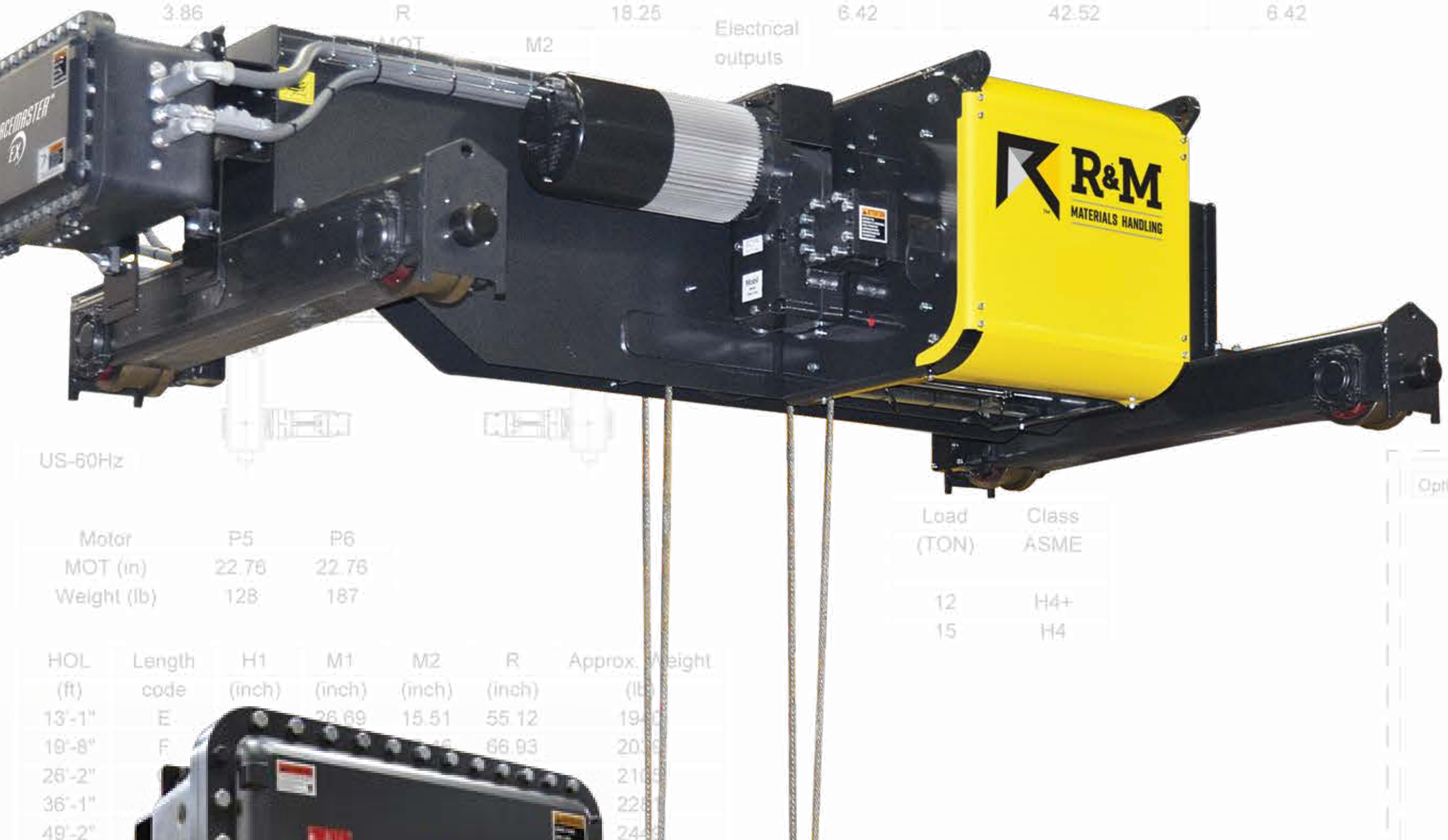
CID1 STANDARD FEATURES:

- 1-20 tons (1000 - 20,000 kg) capacities
- 460 or 575/3/60Hz supply
- Two-speed hoist with 6:1 ratio, contactor control
- Two-speed trolley with 4:1 ratio, contactor control
- Low headroom or normal headroom trolleys
- Single reeved or double reeved
- Intrinsically safe-wired circuitry
- Mainline contactor and control transformer
- Ergonomic and lightweight pendant controller
- CSA C/US approved

SPACEMASTER® EX CRANE PACKAGES

The Spacemaster EX modular crane package includes protective features for the flammable gasses and vapors found in Class I, Division 2 environments.





DESIGNED FOR YOU

Each of our EX hazardous location crane packages are configured specifically to your application. We offer a number of standard features, as well as non-sparking options that are specific to the environments for the Class I, Division 2 explosion proof rating. Some of these additional features include:

- Bronze coated hooks
- Brass wheels
- Brass drop lugs on trolley
- Stainless steel wire rope

STANDARD CRANE FEATURES

- Low headroom, normal headroom and double girder trolleys
- Top or underrunning crane configuration
- 1.5 - 80 ton (1600 - 80,000 kg) capacities
- 460, 575/3/60 or 380/3/50 supply
- ASME H3 or ASME H4 duty (FEM 1Am, 2m, or 3m) rating for hoist
- Single reeved or double reeved
- 2-speed hoist with 6:1 ratio, contactor control
- Low maintenance, sealed hoist brake
- 2-speed trolley with 4:1 ratio, contactor control
- NEMA 4 and 7 protection rating for control enclosure
- CSA C/US approved
- Spark-resistant wire rope as standard (EN 13463-5)

ABOUT SPACEMASTER® EX CRANES:

Spacemaster EX modular crane packages are your one-stop solution for hazardous location cranes. Our packages include pre-engineered components, configured exactly to your specifications. All R&M Spacemaster EX crane package components are suitable for use in Class I, Division 2, Groups B, C and D, and Temperature Class T3 applications.

THE POWER OF LIFTING:

The drive train is the unifying element in R&M's equipment. It's an integrated package of purpose-built gears, motors and controls, designed and made in-house specifically for cranes and lifting motions. This gives our hoists a longer lifetime and excellent functionality. We aim for the highest levels of safety and reliability in our key components.

RISE ABOVE WITH R&M MATERIALS HANDLING, INC.

R&M has been innovating since 1929. With our heritage of innovation, our wire rope and chain hoist products have consistently set the crane industry norms. Our cutting-edge solutions will help you to rise above your material handling challenges. You deserve the best performance possible; let us help you achieve it!

R&M Materials Handling, Inc. may alter or amend the technical specifications identified herein at any time with or without notice.
R&M Materials Handling, Inc. is a proud member of the following organizations:

HEAVY-DUTY WIRE ROPE WINCHES

Heavy-duty lifting requires heavy-duty solutions. The SXL open winch hoist and the PDW process-duty winch are designed for strength and reliability. When you need a rugged hoist with high capacities across a wide range of industries, the SXL and PDW are there to meet the most rigorous lifting demands.

RISE ABOVE™

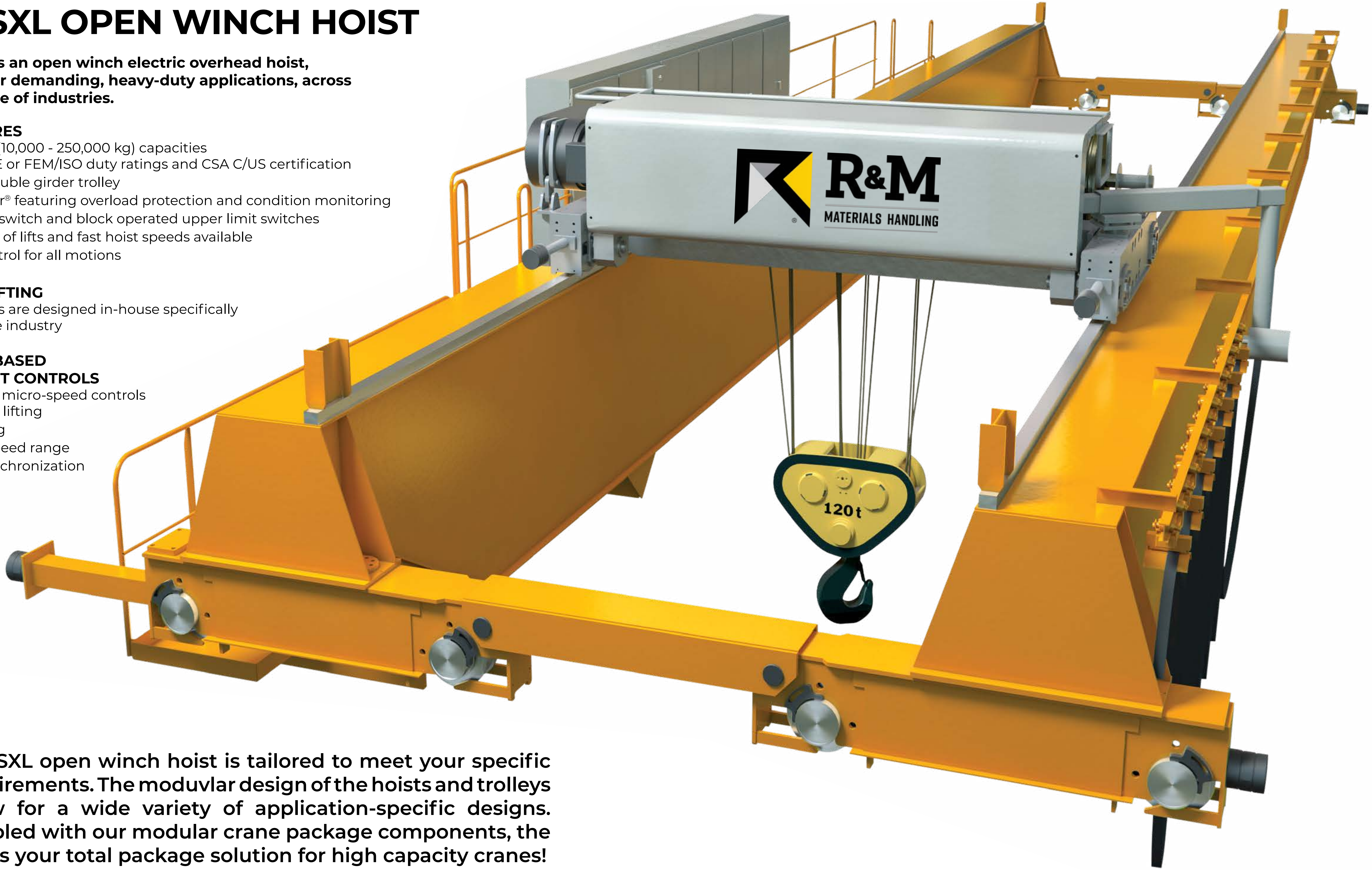
THE SXL OPEN WINCH HOIST

R&M's SXL is an open winch electric overhead hoist, designed for demanding, heavy-duty applications, across a wide range of industries.

SXL FEATURES
- 10 - 250 ton (10,000 - 250,000 kg) capacities
- CMAA/ASME or FEM/ISO duty ratings and CSA C/US certification
- Compact double girder trolley
- HoistMonitor® featuring overload protection and condition monitoring
- Rotary limit switch and block operated upper limit switches
- Long height of lifts and fast hoist speeds available
- Inverter control for all motions

CORE OF LIFTING
- Components are designed in-house specifically for the crane industry

INVERTER-BASED INTELLIGENT CONTROLS
- Inching and micro-speed controls for precision lifting
- Load floating
- Extended speed range
- Hoisting synchronization



The SXL open winch hoist is tailored to meet your specific requirements. The moduvlar design of the hoists and trolleys allow for a wide variety of application-specific designs. Coupled with our modular crane package components, the SXL is your total package solution for high capacity cranes!

THE PDW PROCESS-DUTY WINCH

R&M's PDW hoist is a process-duty winch for a wide variety of heavy-duty applications, such as forging and foundry applications in automotive, steel and metal fabrication industries.

PDW FEATURES

- 6 – 70 ton (6,000 - 70,000 kg) capacities
- Wide range of available lifts and lifting speeds
- Main power supply for 460V-3PH-60Hz or 380-3PH-50Hz
- ASME H4 or H5 duty class (ISO M6-M7) for hoisting
- CMAA Class C-E trolleys (ISO M5-M7) for heavy-duty operation
- Ambient temperatures 41°F - 113°F [5°C - 45°C]
- Inverter duty motors for all motions providing smooth operation and reduced load sway
- Hook latch trigger included
- Shock Load Prevention
- CSA C/US rated

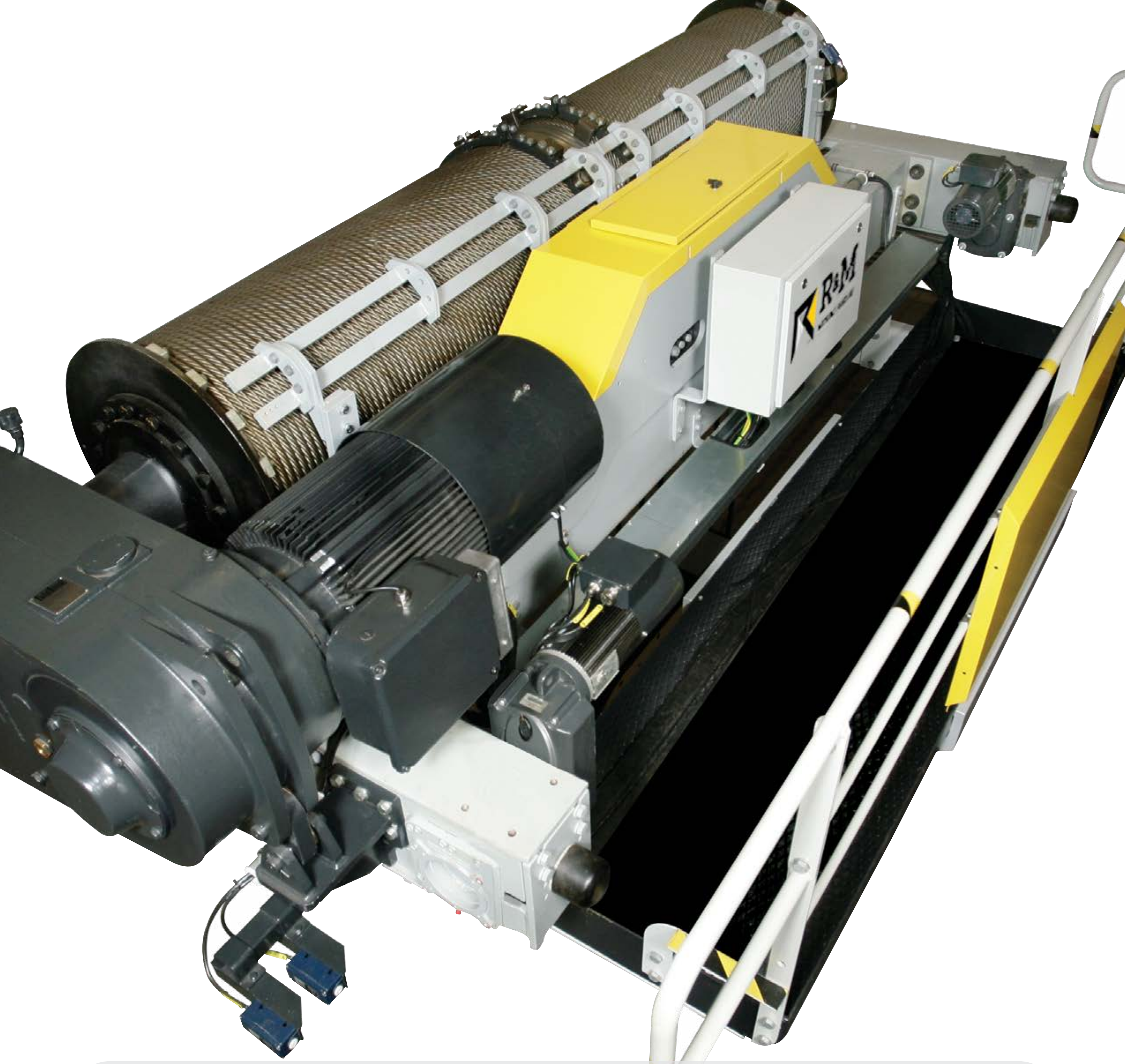
CORE OF LIFTING

- Components are designed in-house specifically for the crane industry

INVERTER-BASED INTELLIGENT CONTROLS

- Extended Speed Range (ESR) with load floating for hoisting
- Allows operator to gain two times the rated speed of the hoist when the load lifted is 20% or less of the rated capacity

The PDW is tailored to meet your specific process-duty requirements. The modular design of the hoist and trolley allows for a wide variety of application-specific designs. Choose from an abundance of special features, such as specialty hooks and provisions for below the hook devices, precision load and sway controls or a maintenance platform. The PDW gives you big impact, in a compact package.



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MATERIAL HANDLING INDUSTRY OF AMERICA Member Company ISA ESTA

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www.rmhoist.com | | Bulletin#SXL PDW 08/27/2025

SPACEMASTER® EX HAZARDOUS LOCATION HOISTS

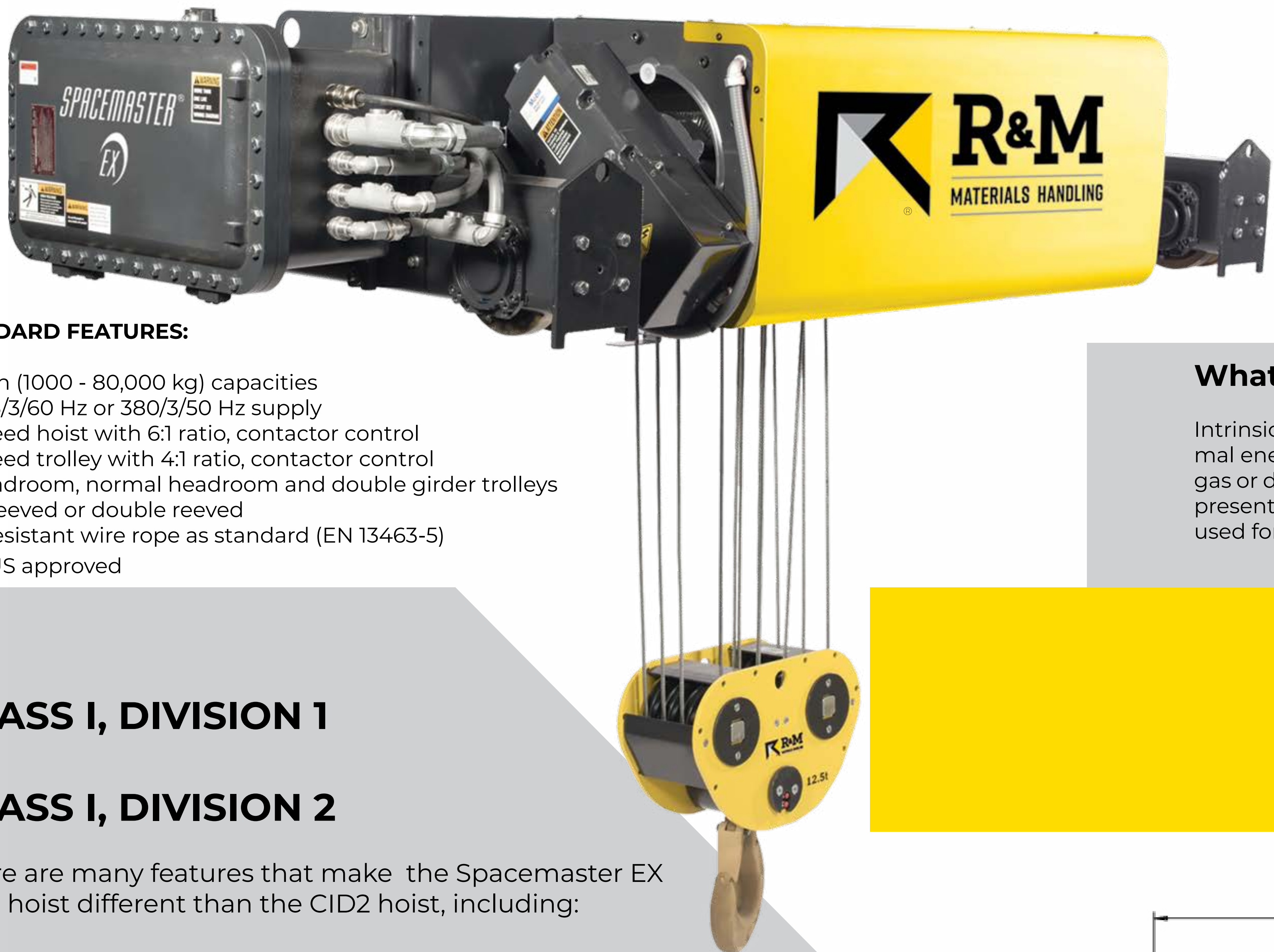
The Spacemaster EX electric wire rope hoist is the hazardous location version of the industry leading Spacemaster SX hoist. Along with exceptional hook approaches and heights of lifts, this hoist features one of the lowest headrooms in the industry. The Spacemaster EX hoists are rated for use in Class I, Division 2 environments with Groups B, C and D, and Temperature Class T3 or in Class I, Division 1 environments with Groups C and D and Temperature Class T4.

RISE ABOVE™

CSA mark shown with adjacent indicators "C" and "US" for Canada and U.S. indicates that the products have been manufactured to the requirements of both Canadian and U.S. Standards

SPACEMASTER® EX CLASS I, DIVISION 2 HOISTS

The Spacemaster EX Class I, Division 2 hoist carries a Class I, Division 2 hazard location rating for Groups B, C and D and with a temperature class T3, and carries an additional marking of Class I, Zone 2 IIB+H2, T3 . Class and Division rated equipment can be used in the appropriate Zone environment as permitted by the National Electrical Code (NEC 505.9 and NEC 505.20) and the Canadian Electrical Code (CEC C22.1).



CID2 STANDARD FEATURES:

- 1 - 80 ton (1000 - 80,000 kg) capacities
- 460, 575/3/60 Hz or 380/3/50 Hz supply
- Two-speed hoist with 6:1 ratio, contactor control
- Two-speed trolley with 4:1 ratio, contactor control
- Low headroom, normal headroom and double girder trolleys
- Single reeved or double reeved
- Spark-resistant wire rope as standard (EN 13463-5)
- CSA C/US approved

CLASS I, DIVISION 1 vs. CLASS I, DIVISION 2

There are many features that make the Spacemaster EX CID1 hoist different than the CID2 hoist, including:

- Different hoisting and traveling motors
- Bigger hoist control panel and new panel fixing plates
- Armored cables instead of flexible conduits for motors
- Different cable entries for motor cables due to armored cables
- Intrinsically safe circuit for control commands and limit switches
- Additional barrier relays because of intrinsically safe circuit
- Additional labels to mark intrinsically safe circuit cables outside the enclosure
- Stainless steel hoist data and warning plates

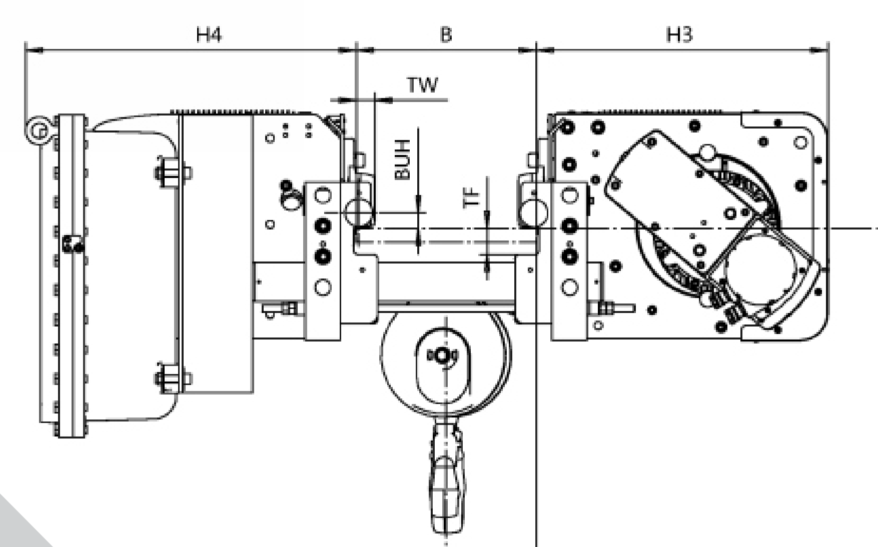
NFPA 70 NEC hazardous location CLASS I *means flammable gasses or vapors may be present ...*	
DIVISION 1	• under normal operating conditions. • as a result of frequent maintenance or repair operations. • as a result of equipment breakdown, faulty operation or failure.
DIVISION 2	• and are normally confined within closed containers when handled or used. • and are normally protected by positive mechanical ventilation. • and the location is adjacent to a Class I, Division 1 location.

What does intrinsically safe-wired mean?

Intrinsically safe electrical systems are designed to limit the electrical and thermal energy in a circuit to a level below what is required to ignite an explosive gas or dust atmosphere. A circuit design considers potential energy that may be present in field devices and interconnecting wiring. Intrinsically safe circuits are used for control commands and limit switches on CID1 equipment.

SPACEMASTER® EX CLASS I, DIVISION 1 HOISTS

The Spacemaster EX Class I, Division hazardous location hoist is certified to meet Class I, Division 1, Groups C & D, and T4 temperature class, and also carries an additional marking of Class I, Zone 1, IIB, T4



CID1 STANDARD FEATURES:

- 1-20 tons (1000 - 20,000 kg) capacities
- 460 or 575/3/60Hz supply
- Two-speed hoist with 6:1 ratio, contactor control
- Two-speed trolley with 4:1 ratio, contactor control
- Low headroom or normal headroom trolleys
- Single reeved or double reeved
- Intrinsically safe-wired circuitry
- Mainline contactor and control transformer
- Ergonomic and lightweight pendant controller
- CSA C/US approved

SPACEMASTER® EX CRANE PACKAGES

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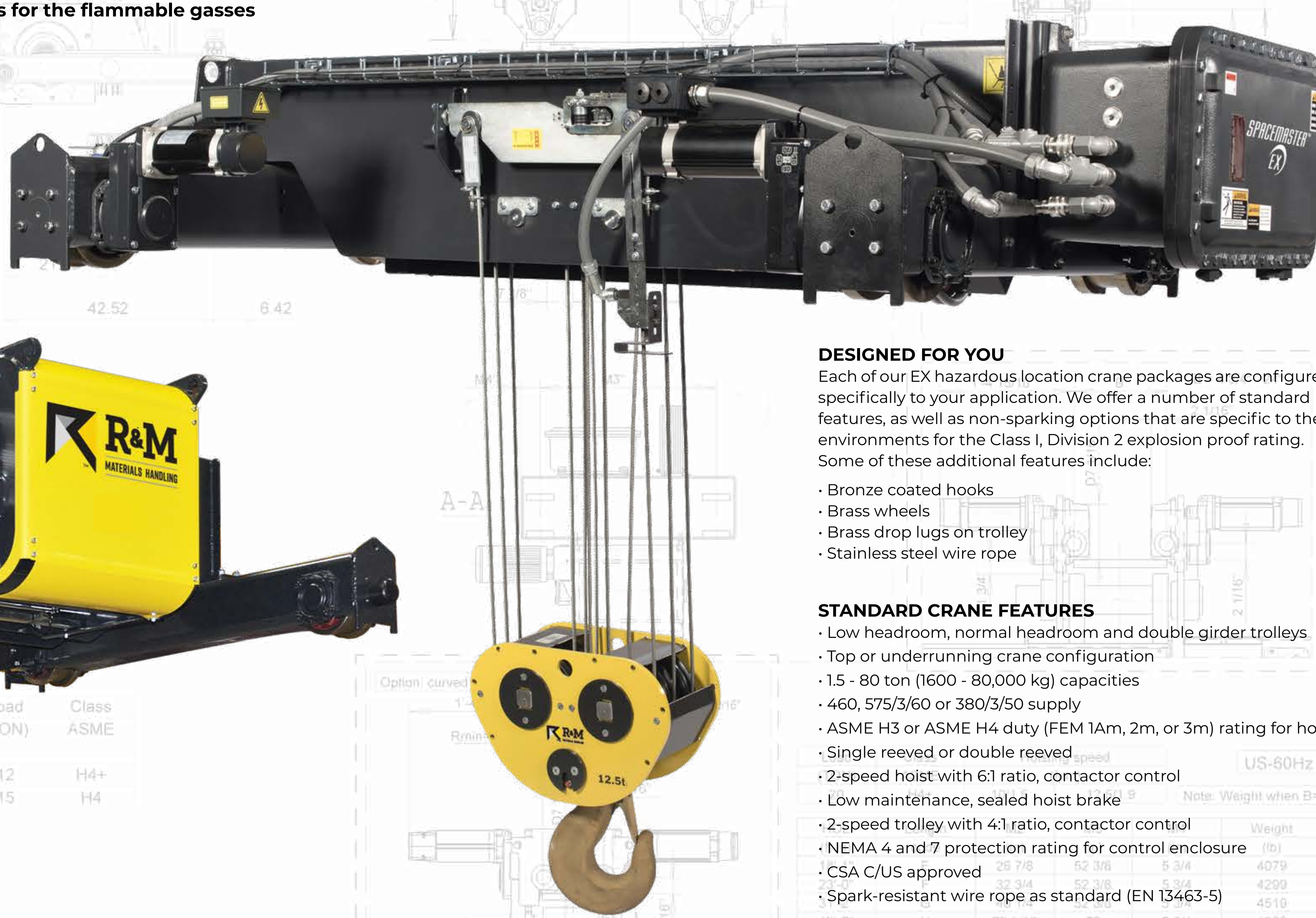

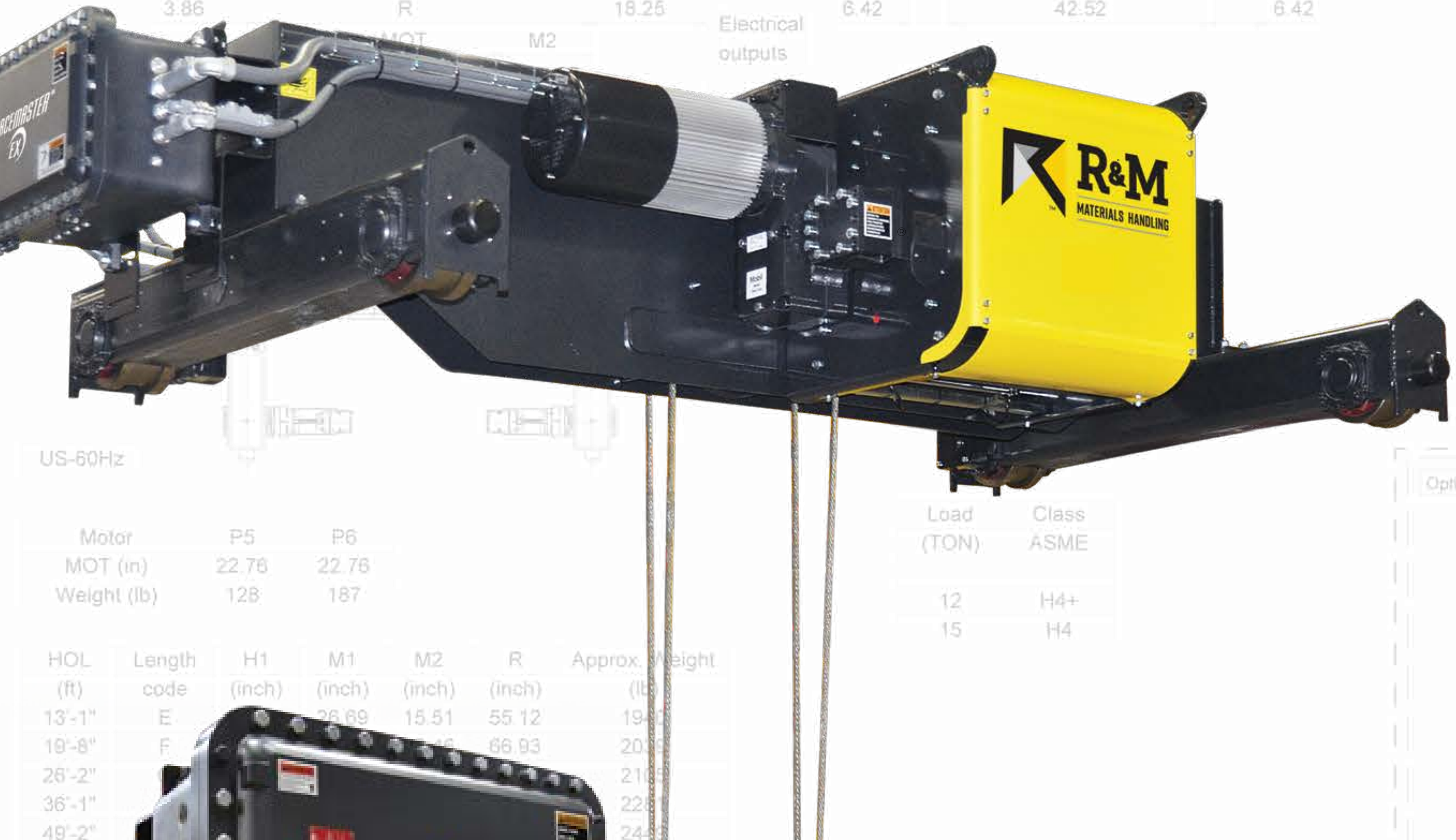

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- Bronze coated hooks
- Brass wheels
- Brass drop lugs on trolley
- Stainless steel wire rope

STANDARD CRANE FEATURES

- Low headroom, normal headroom and double girder trolleys
- Top or underrunning crane configuration
- 1.5 - 80 ton (1600 - 80,000 kg) capacities
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- Single reeved or double reeved
- 2-speed hoist with 6:1 ratio, contactor control
- Low maintenance, sealed hoist brake
- 2-speed trolley with 4:1 ratio, contactor control
- NEMA 4 and 7 protection rating for control enclosure
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THE POWER OF LIFTING:

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RISE ABOVE WITH R&M MATERIALS HANDLING, INC.

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www.rmhoist.com | | Bullet#EX 08/27/2025

LR ELECTRIC CHAIN HOIST

Our new LR chain hoist is built to last, with features that make your job easier. Built smart with the toughness, precision and reliability of our Core of Lifting components, it is designed and tested for more than a million operations over the designed working period of the hoist. As one of the most economical hoists available in the market and a broad range of standard features, the LR hoist brings you industry-leading value to performance ratio. It is everything you want and need for efficiency, safety and productivity.

RISE ABOVE™

LR ELECTRIC CHAIN HOIST

The LR chain hoist features a reimagined motor design that runs cooler so it can run longer, making your business more productive. Innovative technology and optimized mechanics take reliability to a whole new level.

LR ELECTRIC CHAIN HOIST STANDARD FEATURES

- 1/8 - 2 ton (125 - 2000 kg) capacities
- Zinc plated and case-hardened load chain (DAT) 10, 15 and 20 ft (3, 4.5 and 6 m) standard lift heights
  - Additional lift available in 1 ft or 1/2 m increments
- U-shaped bracket suspension connects to a wide range of trolley types for monorails, enclosed track or other fixing points
- Two-speed hoist controls (4:1 speed ratio) - available for various power supplies
- Single speed hoist controls - available for 460V-3Ph-60Hz power supply only
- Upper and lower electrical limit (micro) switches
- Torque-limiting slip clutch device for hoist overload protection
- Plug-in pushbutton pendant with E-stop button
- Indoor use - operational temperature with rated load and speed: -4°F to 104°F [-20°C to 40°C]
- Durable, injection-molded chain container is included

Built-In Overheating Protection

Highly efficient motor fan and large motor fins ensure constant airflow and optimum motor cooling.

Geared Up for Performance

Helical gearing for quiet and efficient operation, gearbox lubrication designed to last for the safe working period of the hoist - enhancing performance and reducing maintenance downtime.

Patented Slip Clutch Design

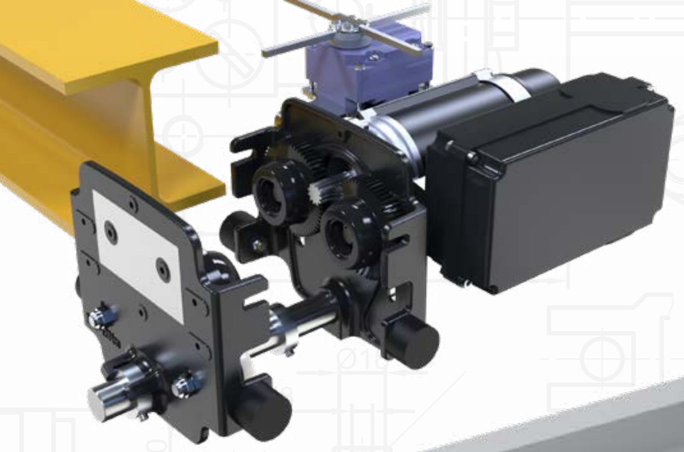
Patented design protects against overloading, enables easy access for service and adjustments and provides outstanding safety and reliability.

Safe & Efficient Braking

Safe brake position prevents hook slippage, and optimized torque reduces shock loading and stress on gearing – leading to long service life for lower-cost operation.

Optimized Chain Drive

Featuring rugged, case-hardened chain for long service life, even in high duty-cycle applications, and 5 or 6 pocket chain wheel options for smooth operations. Safe and reliable upper and lower limit switches included as standard.



WELCOME TO THE NEXT GENERATION OF CHAIN HOISTS

With its rugged, efficient and innovative design, the LR electric chain hoist is built to last, offering outstanding value. Every feature has been engineered to provide efficiency and safety at an economical price point. From our innovative design and rigorous testing, to the ease of installation and operation, the LR electric chain hoist brings more power for your lifting.

- Rugged, innovative design offers outstanding price-to-performance ratio
- Revolutionary U-shaped top suspension bracket design that works as a hook or lug mounted connection and reduces the headroom of the hoist
- Rugged, case-hardened chain and precision-made chain drive system increase chain life and reduce maintenance costs
- Built-in overheating protection prevents damage to critical components

VERSATILITY FOR INDUSTRIAL APPLICATIONS

Compact headroom, a suspension bracket and a range of other features make LR chain hoists easily adjustable to your specific workstation needs.

Manual or Motorized Beam Trolleys

- Adjustable for flange widths of 2.28 in - 12.20 in (58 mm - 310 mm)
- Smooth rolling manual trolley for ease of movement
- Reliable motorized trolley travel in two-speed or inverter driven control types for maximum operator comfort
- Optional travel limit switch available

Installation Made Easy

The LR chain hoist's innovative U-shaped suspension bracket can be used as a lug mounted or hook connection, making installation fast and easy – no special tools are needed.

- Safe and secure locking pins
- Compatible with a large variety of trolleys and fixed suspension applications
- Hoist remains perfectly balanced
- Low headroom design

SAFETY FEATURES

Safety is critical to any industrial operation. Upper and lower limit switches keep you safe and protect your equipment by preventing overtravel of the hook and chain. The LR hoist also comes with order-specific parts, maintenance and safety manuals, so you won't have to guess whether you are following the correct guidelines.

INNOVATIVE PERFORMANCE WITH EASY SERVICE

- Gearbox lubrication that's designed to last
- Plug-and-play electrical components
- Easy-to-access service and inspection points
- Improved chain life with case-hardened DAT chain

HIGH-PERFORMANCE MOTOR

- Heavy duty type 60% ED (CDF) rating (exceeds 30 min. rating) on three-phase motors
- IP55 protection; Class F Insulation
- 4:1 speed ratio for precision load handling

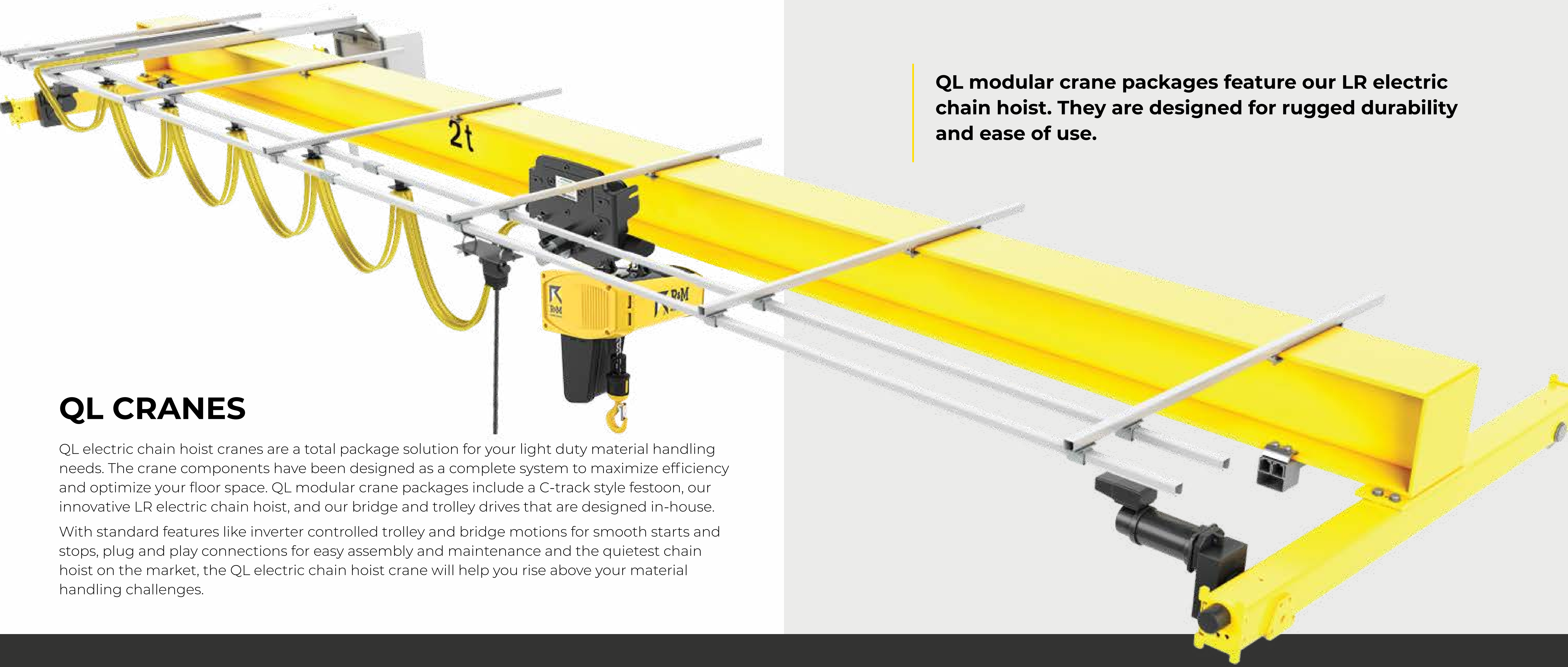
Our LR hoists are factory load tested at 125% of the rated capacity and have a patented slip clutch design that protects against overload.

LOAD WHEEL

LR hoists are fitted with a unique patented chain drive system featuring a precision-cut 5- or 6-pocket load wheel to reduce stress and maintenance needs on the chain. This makes the hoist motion smooth and quiet when operating and works with case-hardened chain to reduce chain wear.

FRAME SIZES & CAPACITY RANGES

Capacity Range - Tons (kg)	Frame Size	Chain Falls
$^1/_8$ - $^1/_4$ ton (125 - 250 kg)	LR02	1
$^1/_4$ - $^1/_2$ ton (250 - 500 kg)	LR02	2
$^1/_8$ - $^1/_2$ ton (160 - 500 kg)	LR05	1
$^1/_2$ - 1 ton (500 - 1000 kg)	LR05	2
$^1/_2$ - 1 $^1/_4$ ton (500 - 1250 kg)	LR10	1
1 $^1/_4$ - 2 ton (1250 - 2000 kg)	LR10	2



QL modular crane packages feature our LR electric chain hoist. They are designed for rugged durability and ease of use.

QL CRANES

QL electric chain hoist cranes are a total package solution for your light duty material handling needs. The crane components have been designed as a complete system to maximize efficiency and optimize your floor space. QL modular crane packages include a C-track style festoon, our innovative LR electric chain hoist, and our bridge and trolley drives that are designed in-house.

With standard features like inverter controlled trolley and bridge motions for smooth starts and stops, plug and play connections for easy assembly and maintenance and the quietest chain hoist on the market, the QL electric chain hoist crane will help you rise above your material handling challenges.

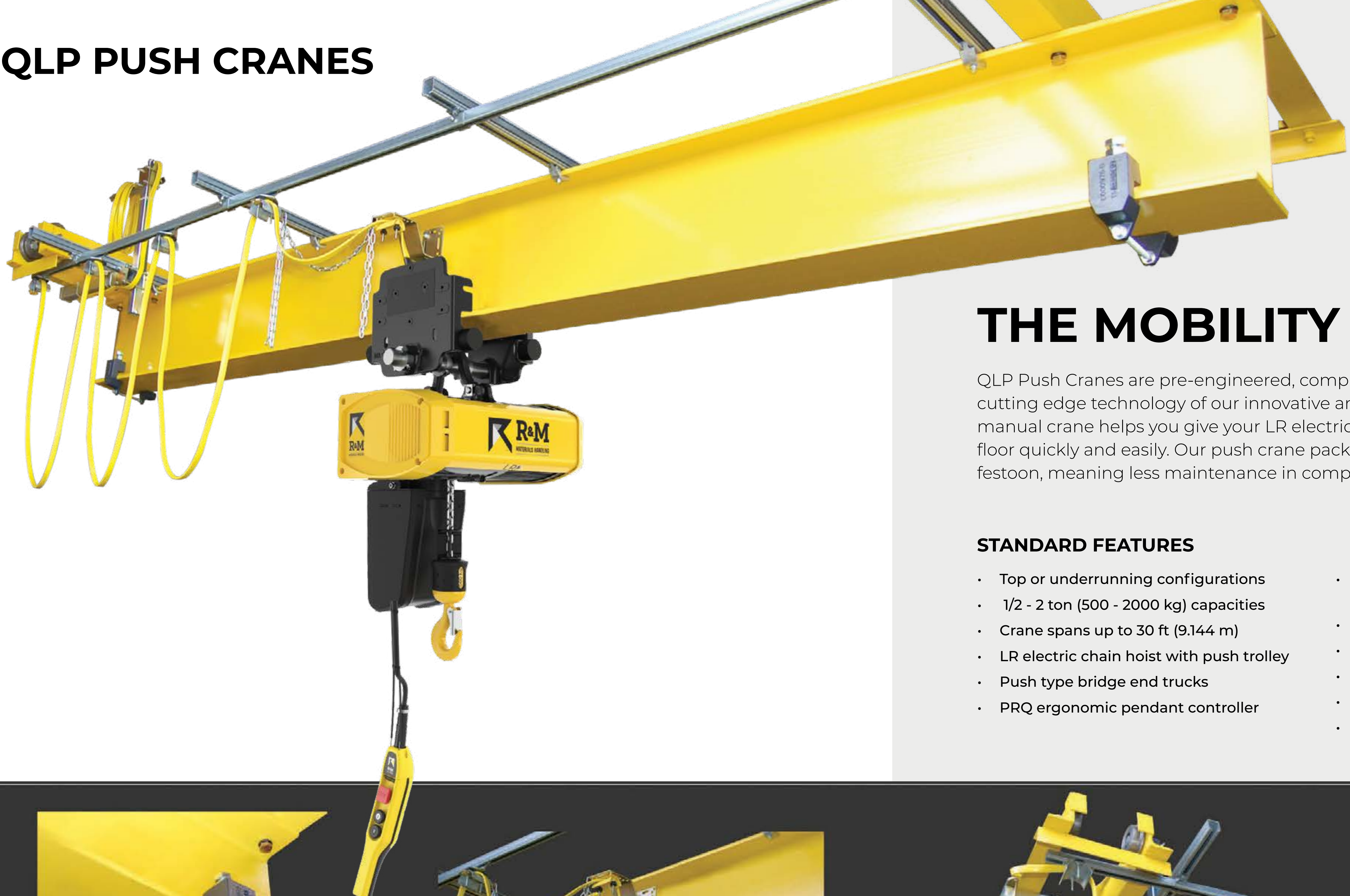
PRECISION LOAD HANDLING

The crane is equipped with all inverter controlled trolley and bridge motions, meaning reduced load swing and better safety. The inverter controls also significantly reduce brake wear on the trolley and bridge. Every LR electric chain hoist comes standard with our innovative slip clutch overload device, protecting you from dangerous accidental overloads. With the slip clutch device, if an operator attempts to lift a load beyond the rated capacity, the load will only be able to lower safely to the ground.

STANDARD FEATURES

- LR electric chain hoist with two-speed hoisting controls
- 1/2 - 5 ton (500 - 5,000 kg) capacities
- Single girder top running or single girder underrunning crane designs
- Standard inverter controls for trolley and bridge for reduced load swing
- Durable C-track festoon system with UL listed cables
- LR electric chain hoist with motorized trolley
- 208, 230, 460 or 575/3/60 power supplies
- Crane spans up to 62 ft [19m]
- PRQ ergonomic pendant controller
- Optional RaCon® ARC radio controller
- CSA C/US approved NEMA 4-3R type/IP55

QLP PUSH CRANES

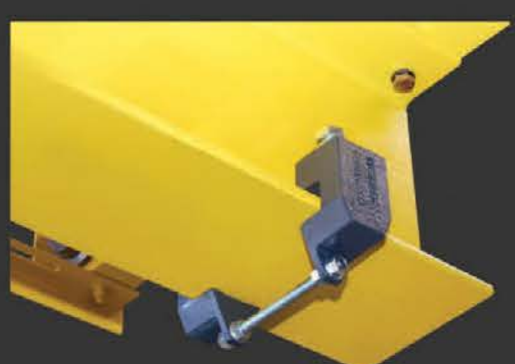


THE MOBILITY YOU NEED

QLP Push Cranes are pre-engineered, complete crane component packages utilizing the cutting edge technology of our innovative and rugged LR electric chain hoist. A push-pull manual crane helps you give your LR electric chain hoist the mobility to cover your shop floor quickly and easily. Our push crane packages are designed with a plug-and-play C-track festoon, meaning less maintenance in comparison to the traditional tag line power supply.

STANDARD FEATURES

- Top or underrunning configurations
- 1/2 - 2 ton (500 - 2000 kg) capacities
- Crane spans up to 30 ft (9.144 m)
- LR electric chain hoist with push trolley
- Push type bridge end trucks
- PRQ ergonomic pendant controller
- C-track style power festoon for industry-leading durability
- Bolt-on end stops for the hoist trolley
- Optional crane tow arm
- IP55 protection rating for indoor service
- Meets CMAA and FEM standards
- CSA C/US approved NEMA 4-3R type/IP55



END STOPS

Included bolt-on end stops for the hoist trolley

TOW ARM

Optional crane tow arm

FESTOON

Comes complete with a plug-and-play power cable, galvanized C-track and cable trolleys

END TRUCKS

Fully assembled end trucks featuring pre-lubricated, antifriction wheel bearings and mandatory safety lugs

LOAD BRAKE GOT YOU DOWN?

R&M's Spacemaster® SX hoist features a squirrel cage type motor with DC brake - the most modern and cost-effective hoisting brake option.

LOAD BRAKE

WHAT IS A MECHANICAL LOAD BRAKE?

The primary function of a mechanical load brake is to provide a "control braking means" to control the lowering speed of the hoist load; the load brake should also be able to hold the load independently of the hoist holding brake. During the first half of the twentieth century, the vast majority of AC motors used on hoisting equipment were "wound rotor motors". Without a control braking means, a load being lowered would continue to accelerate under the force of gravity with little resistance. Unlike a squirrel cage motor, a wound rotor motor does not inherently generate a retarding torque that controls the rate of acceleration of a descending load. The innovative Spacemaster® SX hoist utilizes a squirrel cage motor which is inherently capable of regenerative control braking, while avoiding the many maintenance pitfalls associated with load brake hoists.

LOAD BRAKE PITFALLS

- Increased lifetime costs for the equipment due to increased maintenance and difficult to inspect friction material.
- Load brakes provide no additional safety when a failure occurs beyond the location of the load brake
- Heat and contaminates generated by the load brake will decrease the lifetime of the hoist gear train

ADDITIONAL SAFETY OPTIONS

If redundant safety is required, the Spacemaster SX is available with a secondary holding brake or hoist drum brake option. Both the secondary brake and drum brake options provide redundancy in the event of motor brake failure without compromising the lifetime of the gear train or limiting hoist control options.

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SPECIAL FEATURES FOR R&M HOISTS

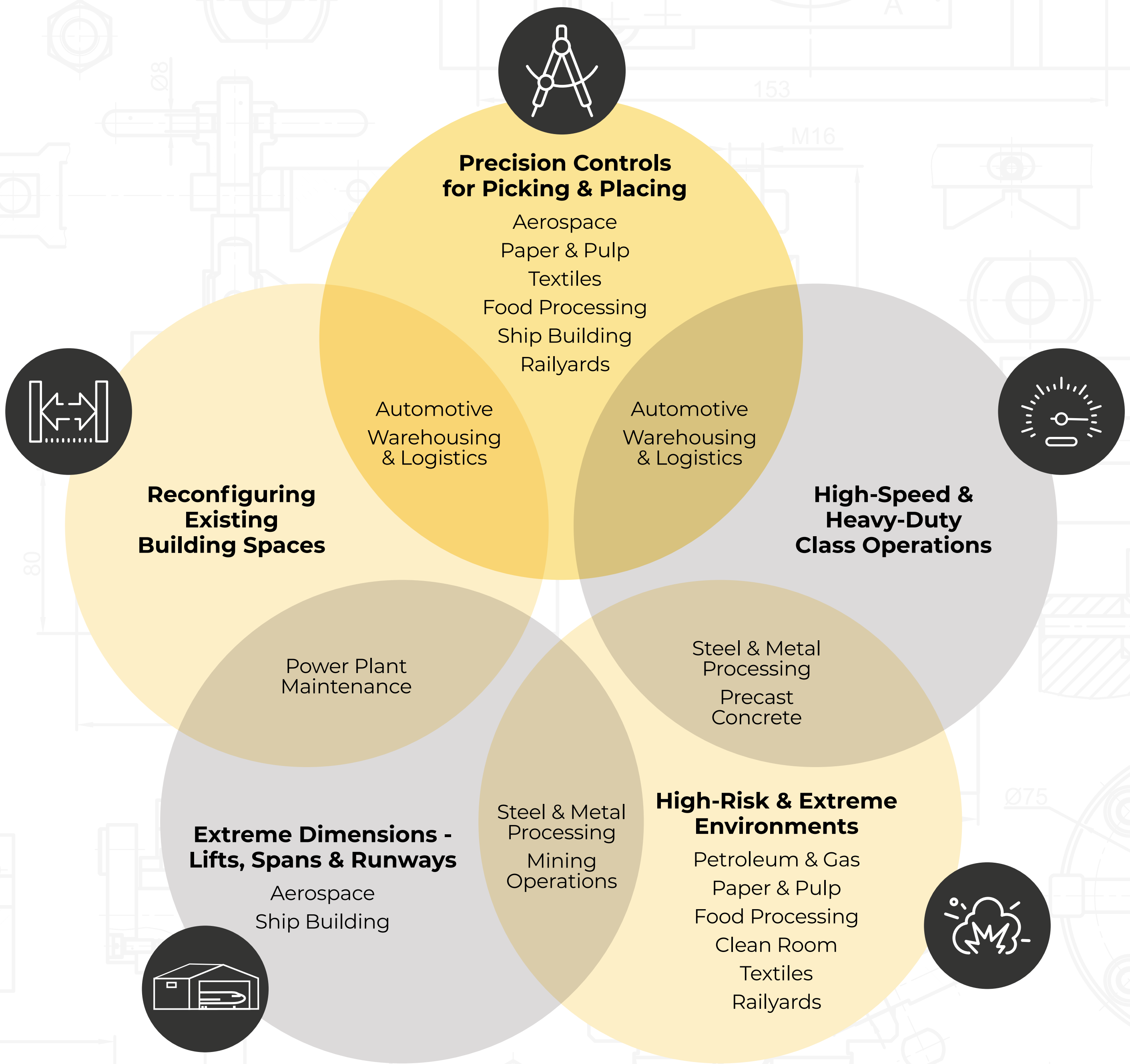
R&M Materials Handing, Inc. offers pre-engineered, complete modular crane packages with advanced hoists, such as our Spacemaster® SX wire rope hoists and high-capacity SXL and PDW open winch hoists. We develop highly engineered, custom solutions for your crane by selecting from our broad range of special features to create a complete system that includes all the functionality you need.

WE'RE PROUD TO HAVE STANDARD OFFERINGS

THAT MANY OTHER LIFTING EQUIPMENT COMPANIES CONSIDER ADD-ON OPTIONS.

However, every industry has its own needs, so we offer special features you can choose for your specific application. We've cross-referenced our special features by application and industry to help you customize your hoist to your needs.

APPLICATIONS & INDUSTRIES



PRECISION CONTROLS FOR PICKING & PLACING

Anti-Sway Technology

ControlMaster® Anti-Sway System can be selected for bridge traveling or for both trolley and bridge traveling. It can be added to cranes with one hoist or two similar hoists. A special encoder on the hoist supplies a signal back to the Height Measuring Unit to accurately predict and counteract the natural pendulum movement of the load automatically. This removes human error and makes the process faster and safer for the operator and the load.

Variable Speed Controls

Infinitely variable speed control gives the user the flexibility to control, accelerate and hold any speed from the minimum to the maximum. The multi-step parameter setting (MS) allows the inverter to be set at two distinct and programmable operating speeds. With either option, the smooth ramp-up and ramp-down prevent load swing and reduce brake wear often associated with contactor controls.

Quick Load / Unload Features

Inverter hoisting controls offer an extended speed range, allowing the unloaded hook to travel faster than the rated hoist speed. This improves load handling when the load is on the hook, and when the hook is empty, it can be quickly moved and reset for the next load.

Load Stabilization

Load floating smart controls suspend the load in a stopped position for a predefined time after the hoisting movement stops without closing the hoist brake. This allows the load to settle completely before taking the next positioning action.

Gentle Material Handling

Inching and micro-speed controls provide a way to approach the load destination with great accuracy through a slow speed range. Depending on the selections and configuration, these motions can be operator-controlled or preset with automation.

Automation Integration

Closed-loop inverter controls, along with smart features, allow for full automation of standard picking and placing processes to save time, increase production capacity and improve operator safety. Automation removes human error from repetitive, precise movements.

Below-the-Hook Devices

R&M's equipment can integrate with specialty below-the-hook devices such as magnets, shakers, rotating blocks, spreaders and vacuum lifters.

Load-Turning

Load-turning features help increase speed and safety during a load-turning process, such as die-flipping. This feature protects the hoist from side pull outside of the allowable range, improving operator safety and reducing the risk of costly equipment damage.

Advanced Positioning Systems

Smart features and automation provide a way to approach the load destination with great accuracy and minimized operator interaction. This leads to improved safety and increased productivity.

HIGH-SPEED & HEAVY-DUTY CLASS OPERATIONS

Quick Load / Unload Features

Inverter hoisting controls offer an extended speed range, improving load handling when the load is on the hook. This reduces downtime and increases productivity.

Robust Construction

The Process Duty Winch (PDW) provides excellent performance in process duty applications for Class D and Class E duty cycles. It is available for loads between 6 - 70 tons (6,300 - 70,000 kg) and with a wide range of hoist speeds and lifts.

Protecting Your Investment

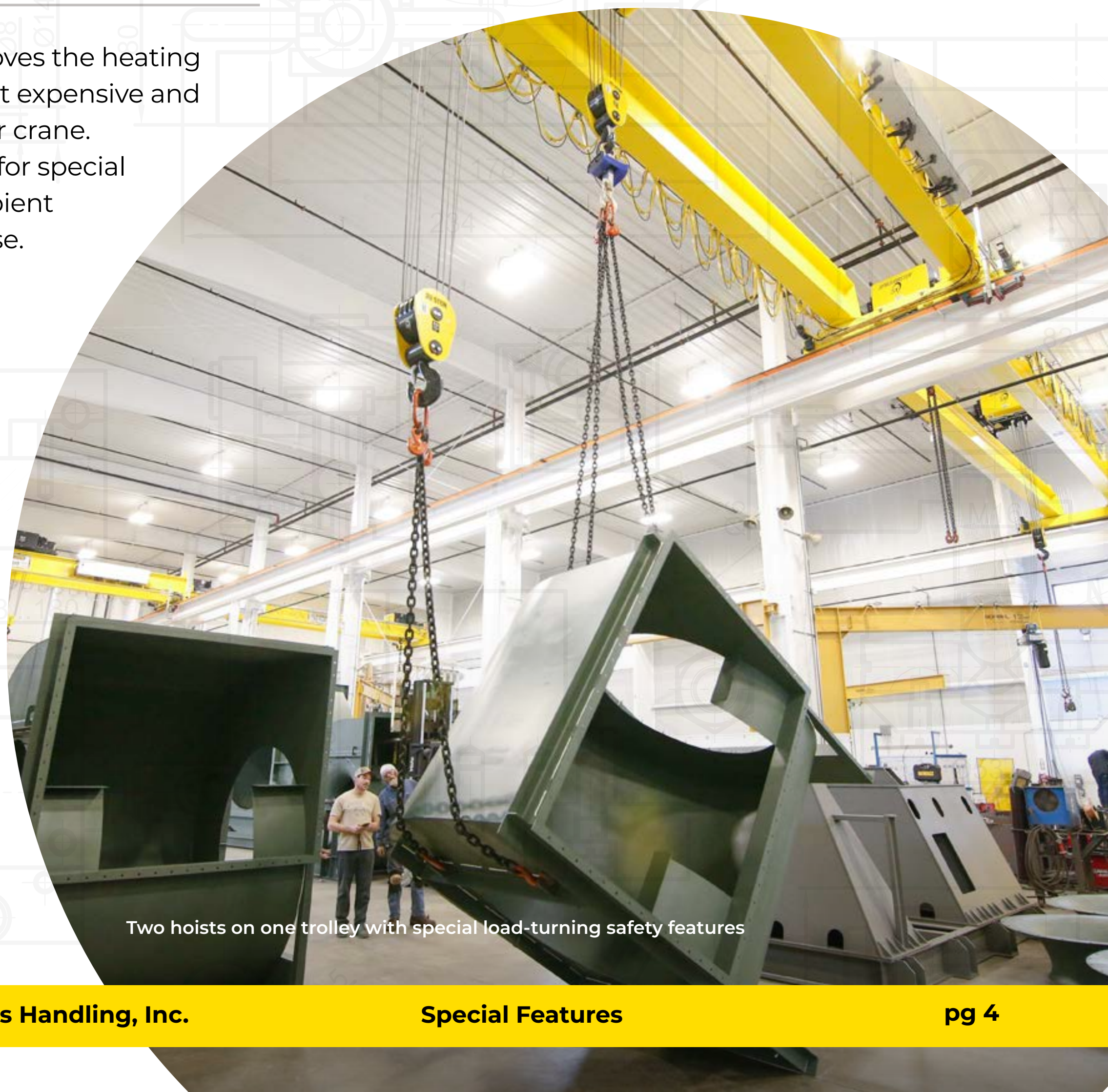
Class H motor insulation improves the heating capacity of the motor, the most expensive and critical component of a hoist or crane. Class H motors are often used for special applications, such as high ambient temperatures or heavy-duty use. Keeping them insulated in these environments is critical to their performance.

Condition Monitoring

HoistMonitor® Enclave is added to all R&M wire rope hoists for overload protection, hoist duty class monitoring, shock load prevention, improved safety and lengthening the life of the hoist.

Remote Monitoring to Reduce Downtime

The HoistMonitor Enclave, coupled with our gateway modem, provides wireless connections for remote monitoring of parameters, which can prevent breakdowns and allow for predictive maintenance and troubleshooting without additional service visits.



Two hoists on one trolley with special load-turning safety features

HIGH-RISK & EXTREME ENVIRONMENTS

Critical Lifts

A variety of smart controls and safety features are added to the hoist and crane components when operators must come in close contact with the load as a part of the standard operating procedure. Depending on the application, these could include true vertical lift, anti-sway, micro-speeds and additional redundant limits and/or brakes.

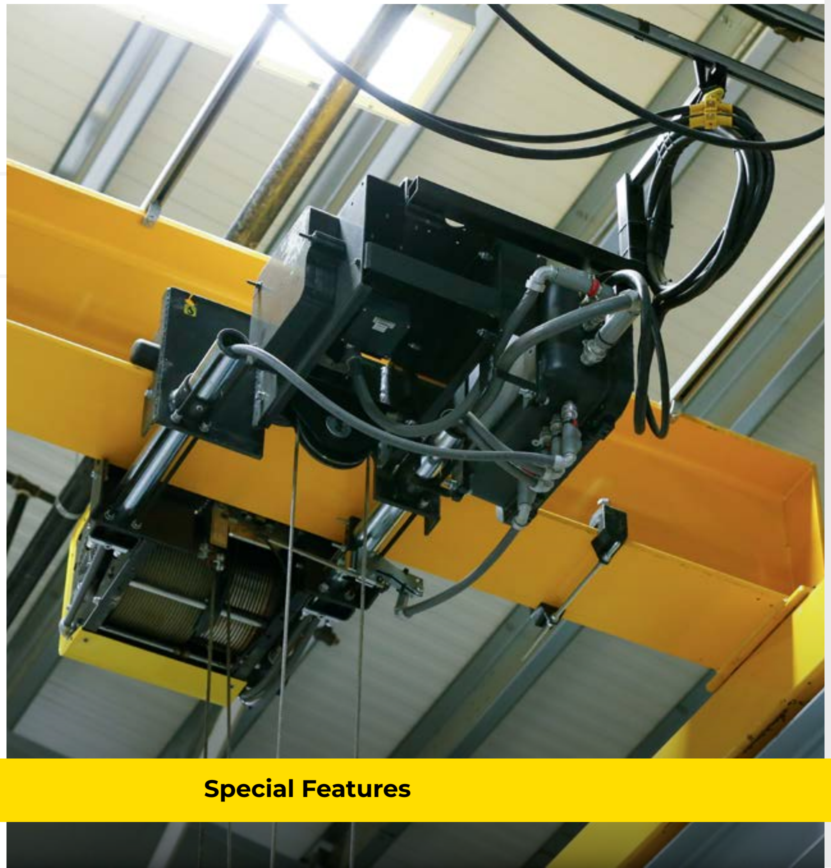
Redundant Braking

A second hoist brake can be added to the SX2, SX3, SX4 or SX5 hoists for additional safety. Both brakes open simultaneously, but the closing of the second brake is momentarily delayed, avoiding rapid stop and excessive brake wear. With this momentary delay, the second brake acts as a holding brake that reduces the risk of falling loads that could damage the load and injure employees. For larger hoists (SX5-7, SXL and PDW), a hoist drum brake can be added for additional safety.

Explosion-Proof and Non-Sparking

R&M's Spacemaster EX hoist is available with a Class I, Division 1 or Class I, Division 2 rating. Non-sparking options include bronze-coated hooks, brass wheels, brass drop lugs on the trolley and stainless steel wire rope. Additional control options, like an explosion-proof radio control unit, improve crane safety and productivity by allowing the operator a greater range of movement while maintaining clear sight lines to the load.

Equipment rated for Class I, Division 2 hazardous locations in a municipal power plant



Safety Features

The run and fault supervision feature checks for any active faults from those functions that could be activated. If a fault exists, lifting, lowering or both (depending on the faulted function) will be disabled until the fault is cleared. If no defaults are detected, then lifting or lowering will be allowed. Anti-collision devices prevent cranes on the same runway from approaching each other at a high rate of speed.

Dust and Moisture Resistance

IP66 protection is added to the motors and electrical panels to prevent dust and moisture from entering the components. The bottom side drum cover provides additional protection from dust and moisture coming from below the hoist. Rail sweeps are added if the trolley rail can be covered with dust or debris, even indoors, like in a paper mill or steel fabrication facility.

Weather Resistance

Rain covers protect the hoist and traveling machinery from overhead precipitation. IP66 protection includes a specially sealed motor, brake and electrical panels for humid or dusty conditions. Rail sweeps are added if the trolley rail can be covered with snow, dust or debris, and special paint thicknesses can be added to protect the hoist body from corrosive atmospheres, like in offshore applications. Anti-jump catches can be added to the top running double girder trolley to prevent the trolley from leaving the rail in cases of high winds.

High Altitude

When operating at high altitudes > 3,300 ft (1000 m), the hoist's capacity is reduced, so the hoist motor, trolley and bridge motors will need to be upsized. R&M's wide range of available motor and gear pairs can often accommodate this need without having to unnecessarily upsize the mechanical components.

Corrosive Environments / Galvanizing

Special paint and special paint thicknesses can be applied to better protect against corrosive vapors, such as hydrochloric acid, sulfuric acid or sulfur dioxide, rising from preparation pools. Specialty cast iron motor housings are also available for galvanizing applications.

Clean Room Compatibility

Overhead Lifting Information (OLI) lets you view the HoistMonitor data remotely through a wireless connection, so you don't have to enter the clean room to check on stats, such as the condition of the hoist and its motor, brakes or contactors. Inverter-based intelligent features help with load handling and positioning, minimizing contamination possibilities.

Extreme Temperature Protection

Standby heating for hoist, bridge and trolley motors prevent condensation from building up on the motor windings during prolonged periods of idleness. Special metal types for some structural components, the elimination of plastic parts and synthetic lubricants are also applied in extremely low-temperature applications (below -40° F (-40° C)). Fan and air-conditioned cooling types may be used in very warm climates to protect the electronic components.

EXTREME DIMENSIONS - LIFTS, SPANS & RUNWAYS

Cantilever Cranes

Bridge girders on an underrunning crane can be engineered with a cantilever on either or both ends to extend the span for the application while utilizing the existing runway structure.

Extended Reach

Specialty hoist drums are available for some double girder and normal headroom hoists to achieve longer than standard lift heights. In some applications, a cost-effective alternative is to increase the reach of the hoist by adding additional rope to the drum and extending the headroom of the hoist.

High-capacity crane with SXL hoist for metal fabrication

Multiple Runways

Multiple runways on an underrunning crane allow for very long open spans to be achieved in lighter capacity applications without the use of large and costly box girders. If you have a multiple-runway application, consider increased travel speeds to cover the span or runway effectively and use RaCon® radio controllers for optimum floor coverage and ease of operation.

Long Spans

Bogie end trucks are recommended in long-span applications. Because the four-wheel design disperses the wheel loads, smaller rail sizes can be used. Additionally, bogie end carriages make installation quicker and safer in long-span applications. Box girders are ideal for carrying heavier loads on longer spans. R&M's design tool specifies a torsional style box beam, which optimizes the usage of steel, end truck size and loading on the runway.

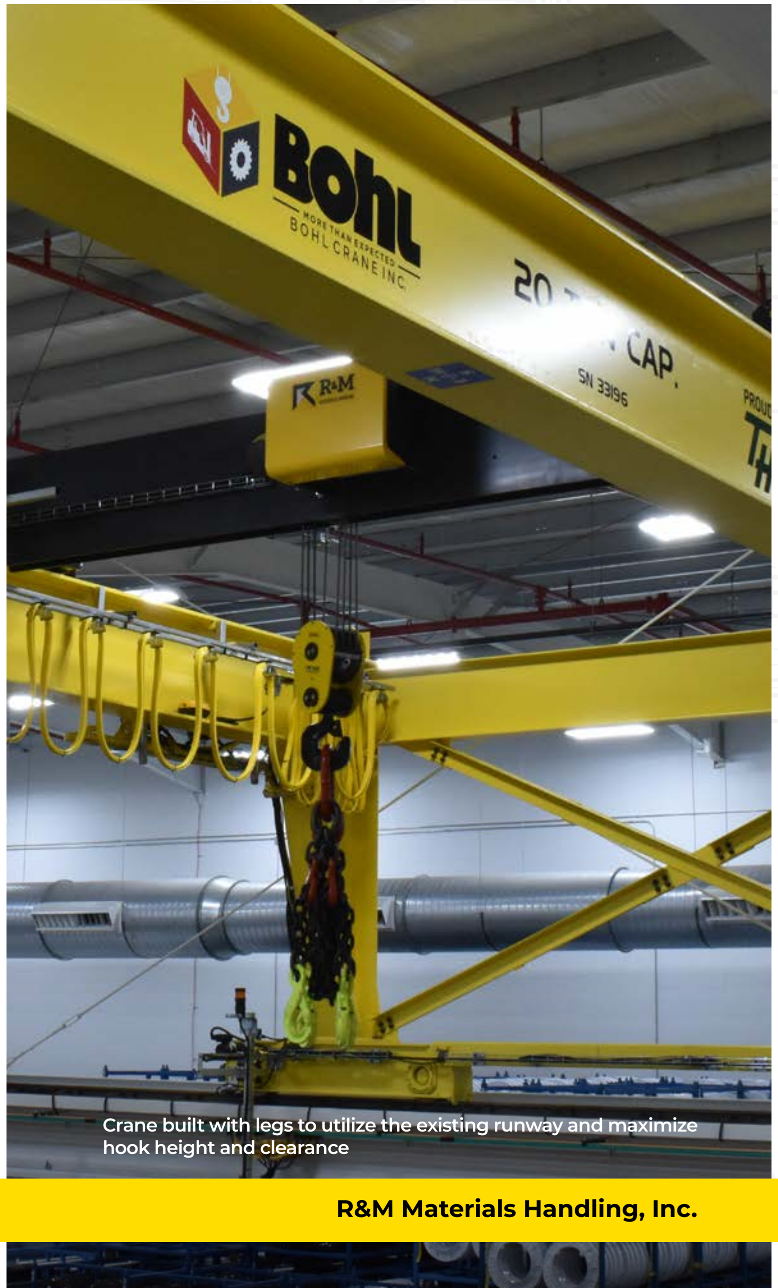
Heavy Usage

Heavy-duty wire rope is recommended when the hoist is used at constant near capacity and has a high duty cycle. Class H trolley motor insulation improves the motor's heating capacity, which is often needed in heavy-duty applications. Another option is to de-rate a larger hoist, effectively reducing the calculated duty cycle for the application. Inverter controlled hoisting also improves the hoist lifetime by reducing wear to the rope and hoist brake, while also offering improved operator safety and performance.

RECONFIGURING EXISTING BUILDING SPACES

Space Optimization

R&M's proprietary design software utilizes your building's existing dimensions to optimize the hook height and overhead clearance of the crane. The low headroom Spacemaster SX hoist's unique drum design also helps to improve trolley end approach. This customized approach to crane design means improved floor coverage and a maximized space within the building envelope.



Crane built with legs to utilize the existing runway and maximize hook height and clearance

Flexible Trolley Configurations

A variety of specialty designs are available as an alternative to our standard normal and double girder trolley options. Normal headroom trolleys can be turned 90 degrees or provided as a lug mount connection to accommodate space requirements. Double girder trolleys are also available in both high and low connection types, with a wide range of trolley gauges to accommodate the replacement of older hoists and trolleys, which were often larger in size. Our flexible trolley gauge hoist allows for quick delivery of replacement double girder hoists with custom gauges.

Gantry and Stooled Up Cranes

Gantry cranes offer flexible configurations that allow them to be used in multiple areas. Whether double-legged or single-legged, gantry cranes can be used for a wide range of capacities and lifts. They are ideal for moving heavy loads to and from different areas, such as from room to room or indoors to outdoors. In cases where an existing runway is being re-used for a new crane, a stooled up design employs many of the same design principles as a gantry, while maximizing headroom and lifting capabilities on the existing runway.

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At R&M Materials Handling, Inc., we want to ensure you get exactly what you need for your application and industry. Our goal is to deliver products that exceed expectations. We leverage our 90 years of experience to ensure your hoist has all the functionality you need.

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